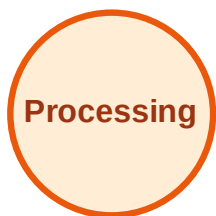


BFS Traversal

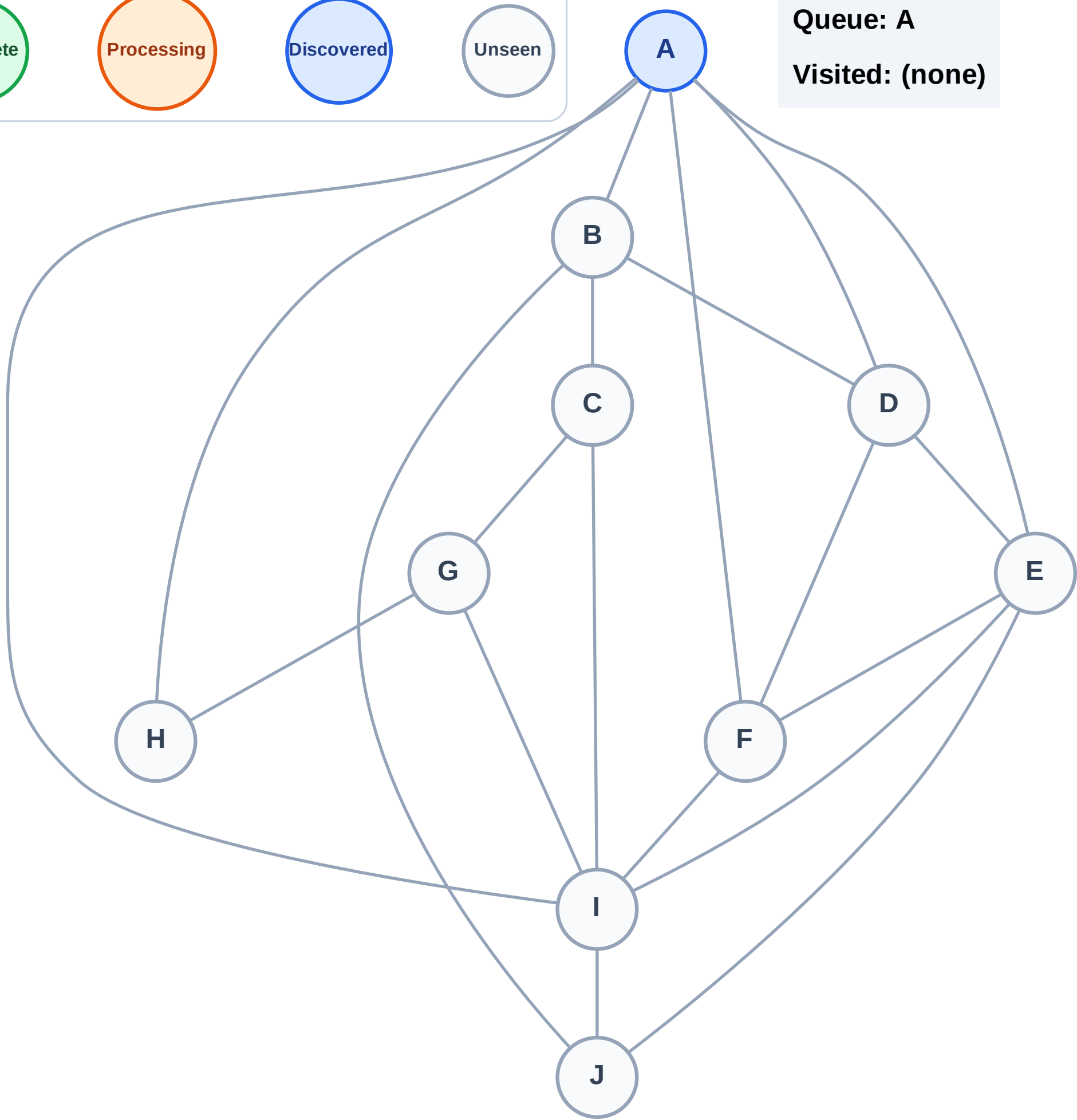
Step 1: Start at A

Legend



Queue: A

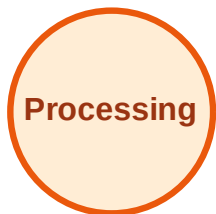
Visited: (none)



BFS Traversal

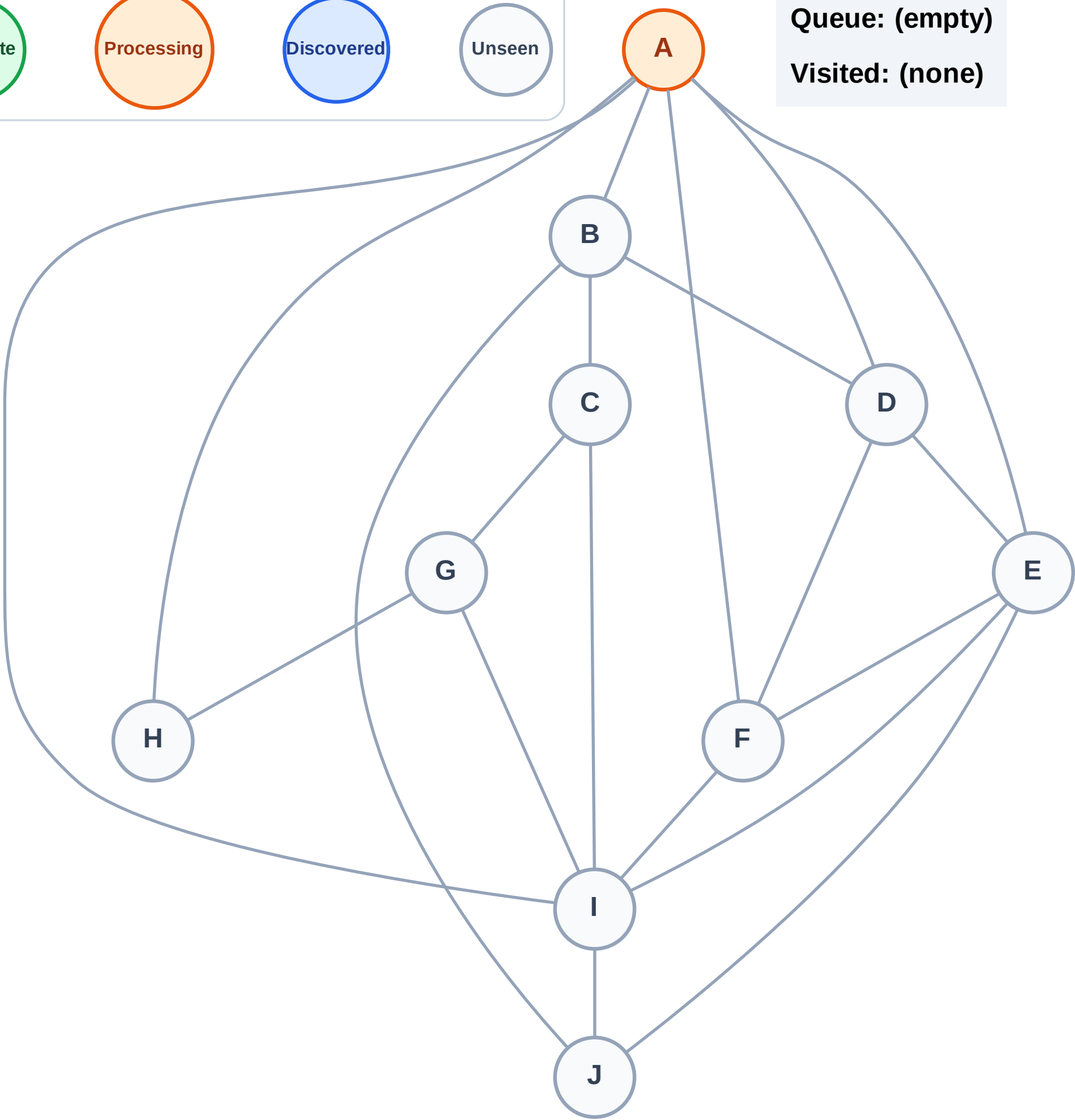
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

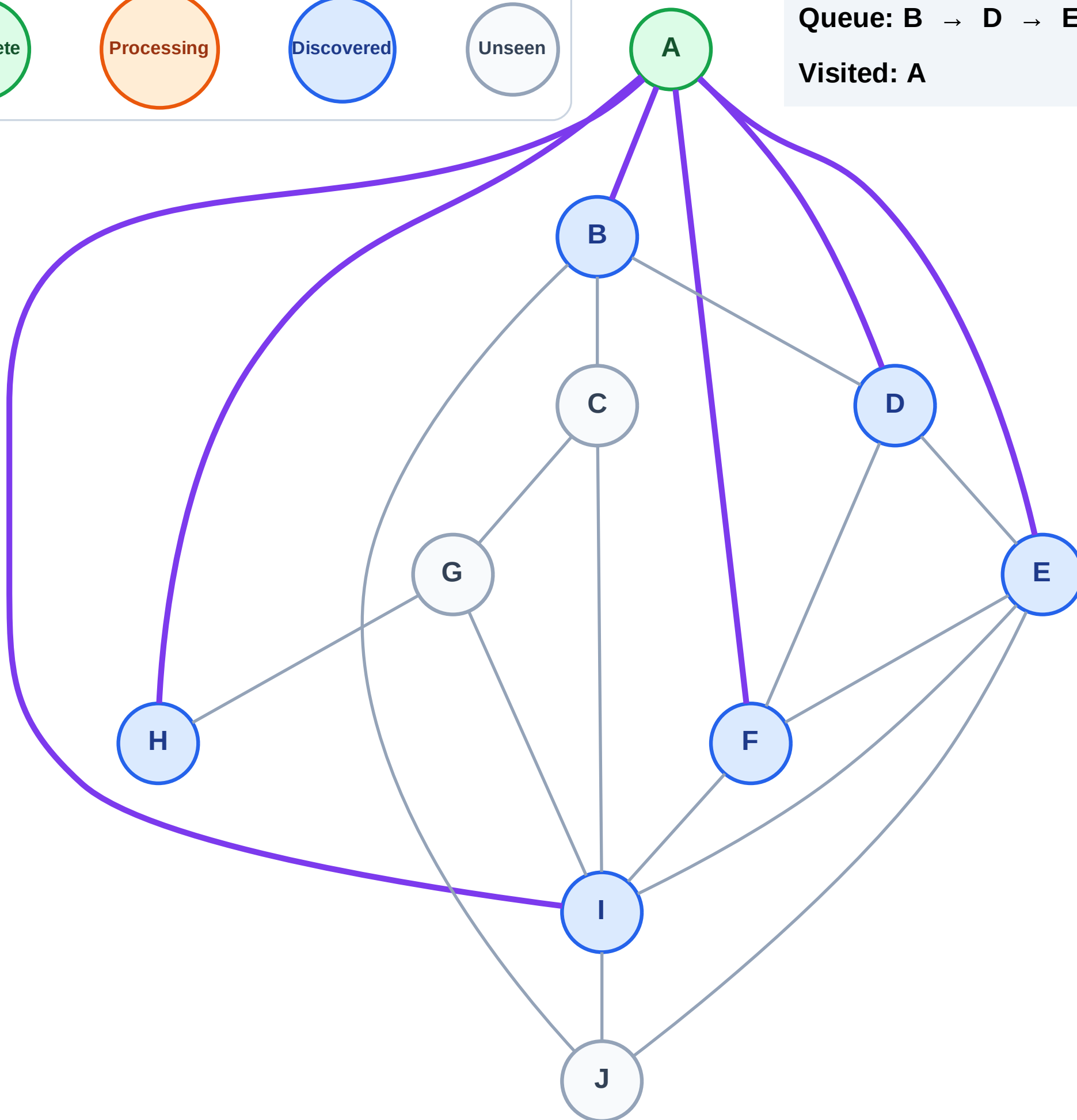
Step 3: Discover B, D, E, F, H, I from A

Legend



Queue: B → D → E → F → H → I

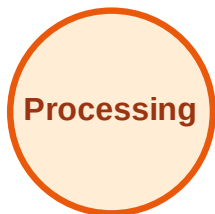
Visited: A



BFS Traversal

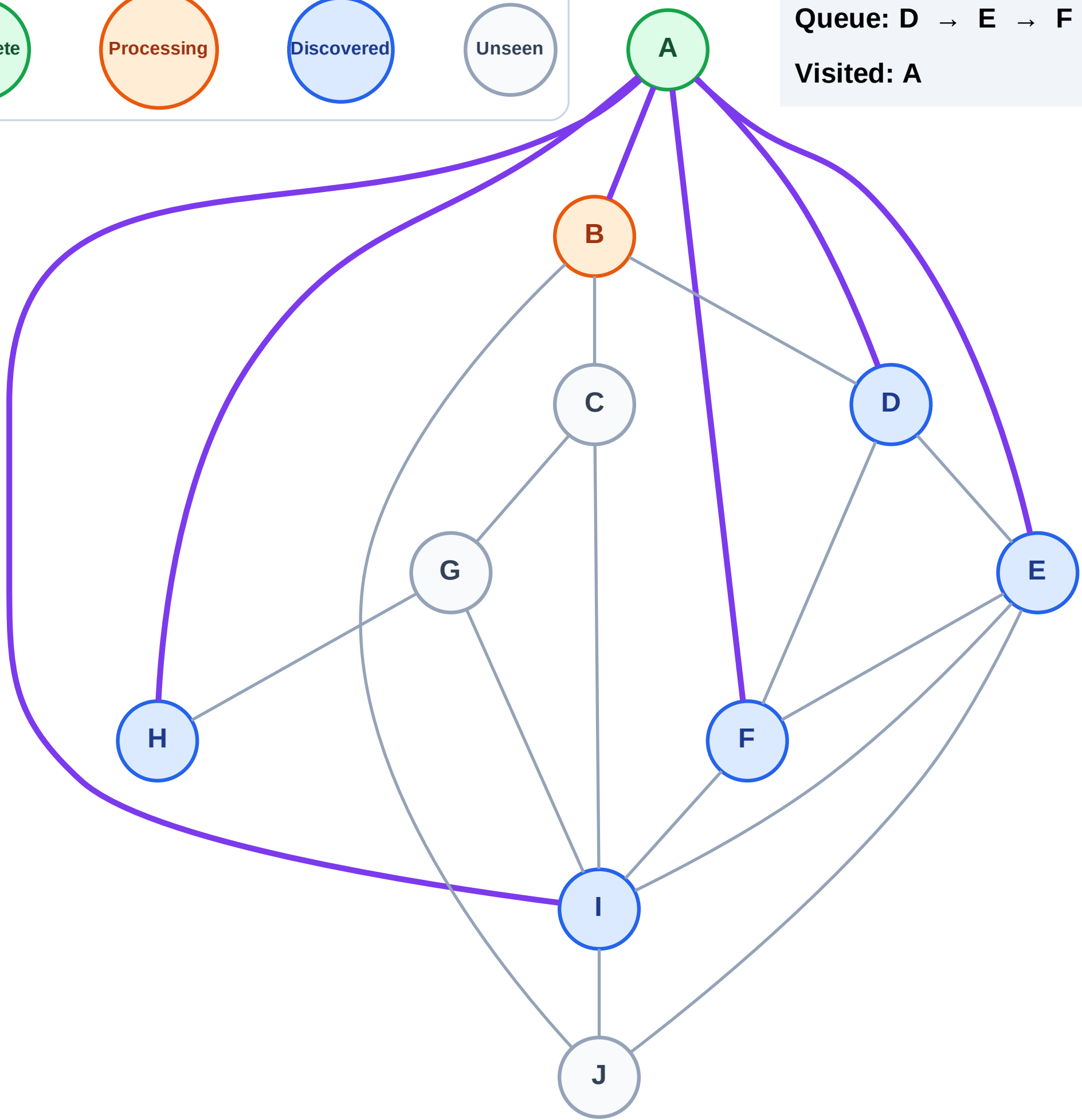
Step 4: Process node B

Legend



Queue: D → E → F → H → I

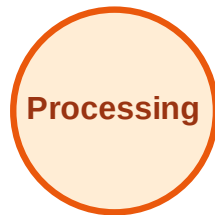
Visited: A



BFS Traversal

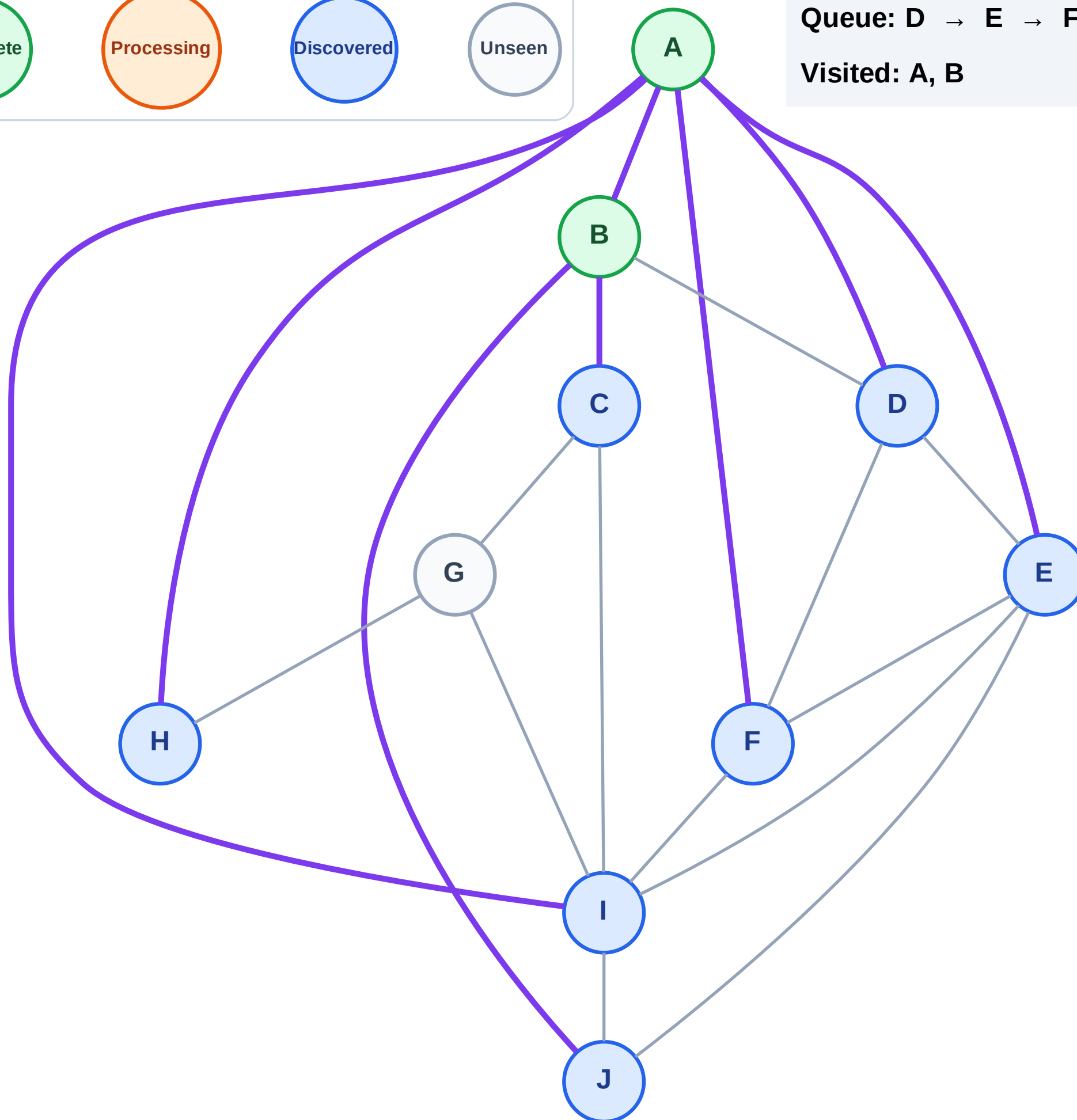
Step 5: Discover C, J from B

Legend



Queue: D → E → F → H → I → C → J

Visited: A, B

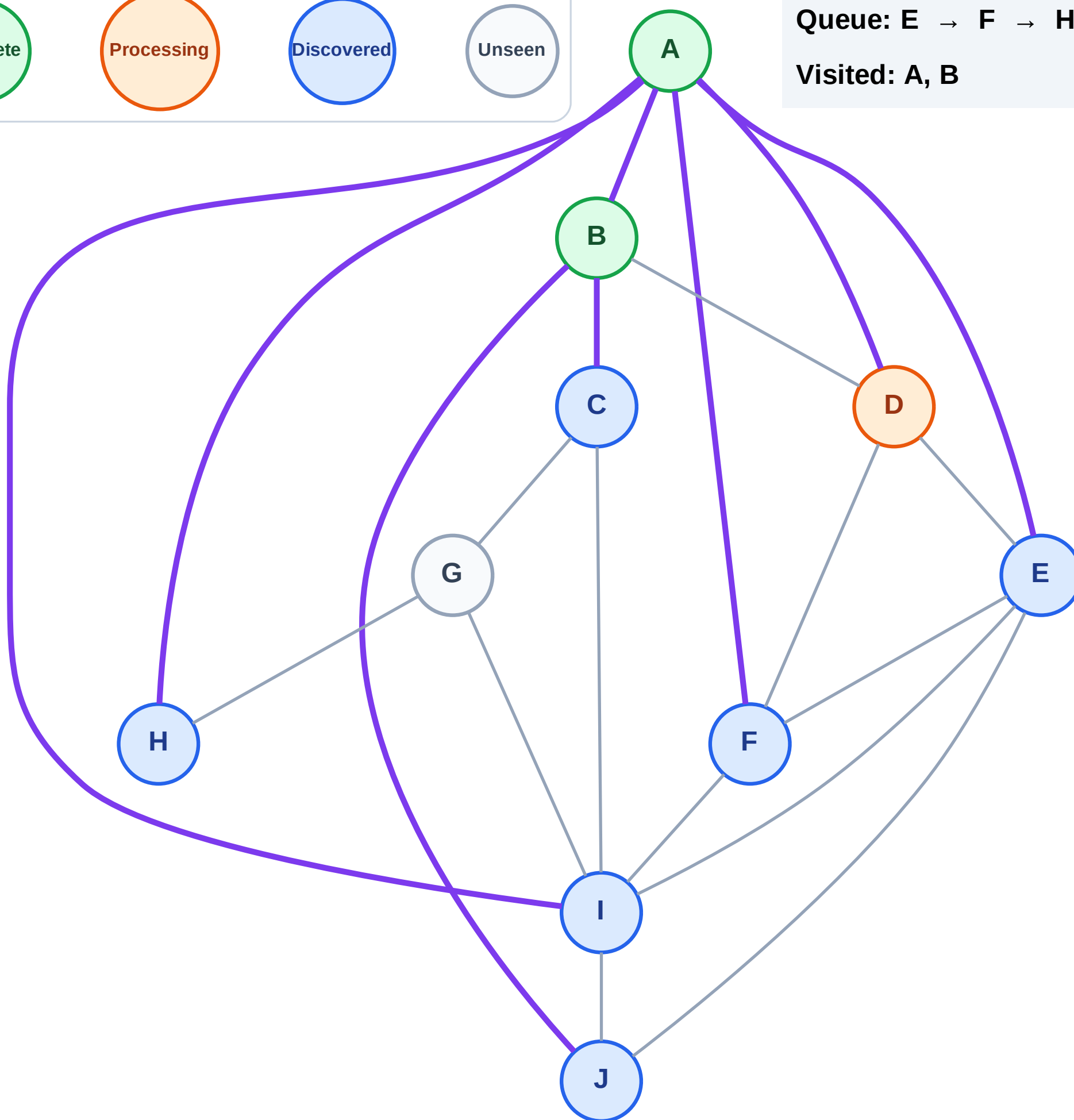
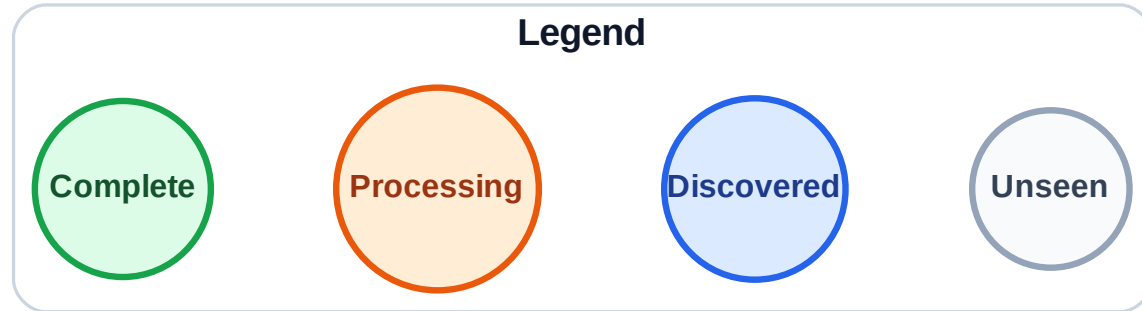


BFS Traversal

Step 6: Process node D

Queue: E → F → H → I → C → J

Visited: A, B



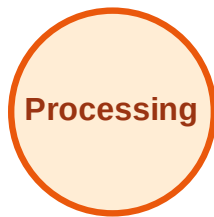
BFS Traversal

Step 7: No new neighbors from D

Legend



Complete



Processing



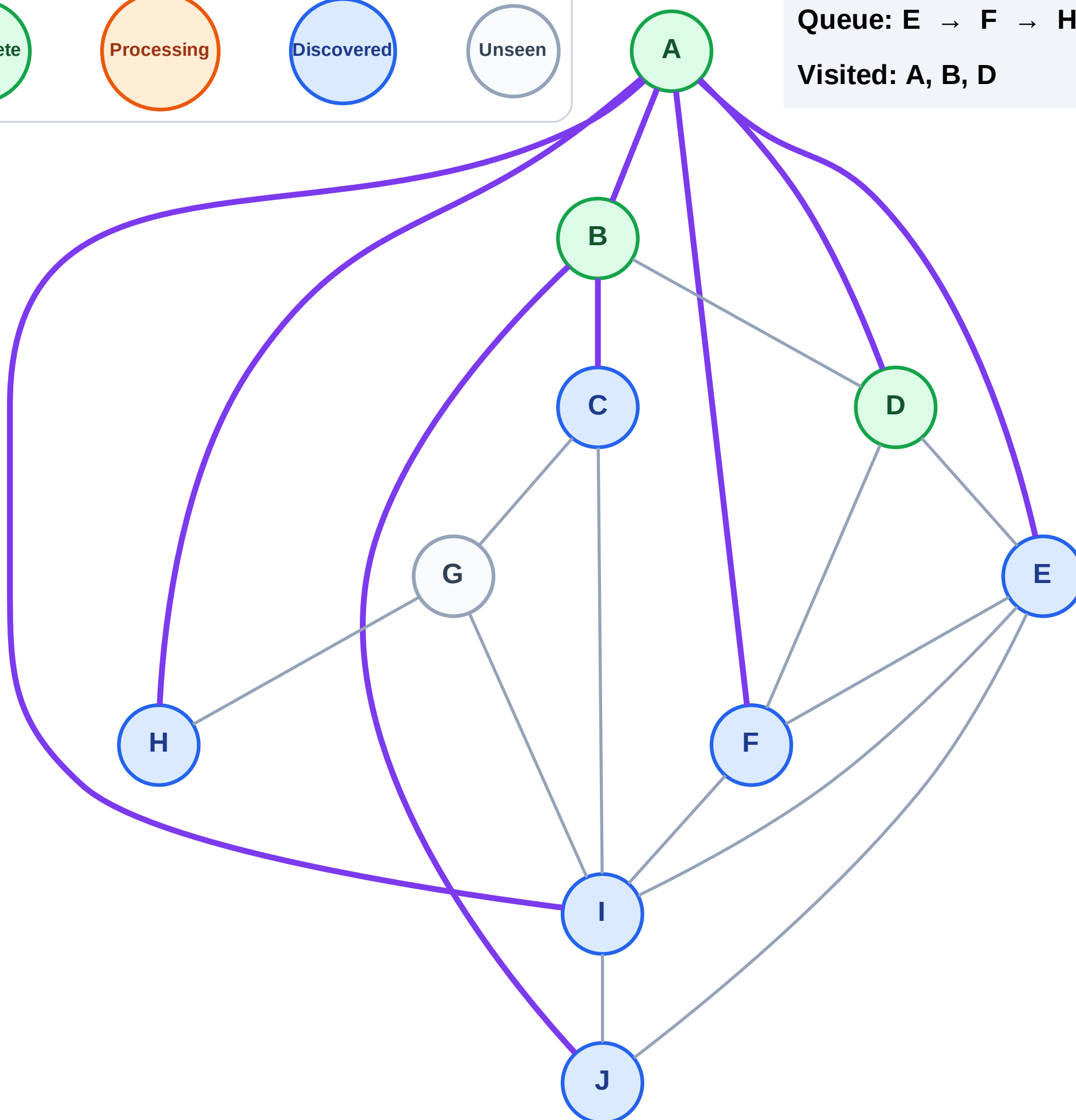
Discovered



Unseen

Queue: E → F → H → I → C → J

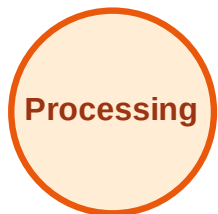
Visited: A, B, D



BFS Traversal

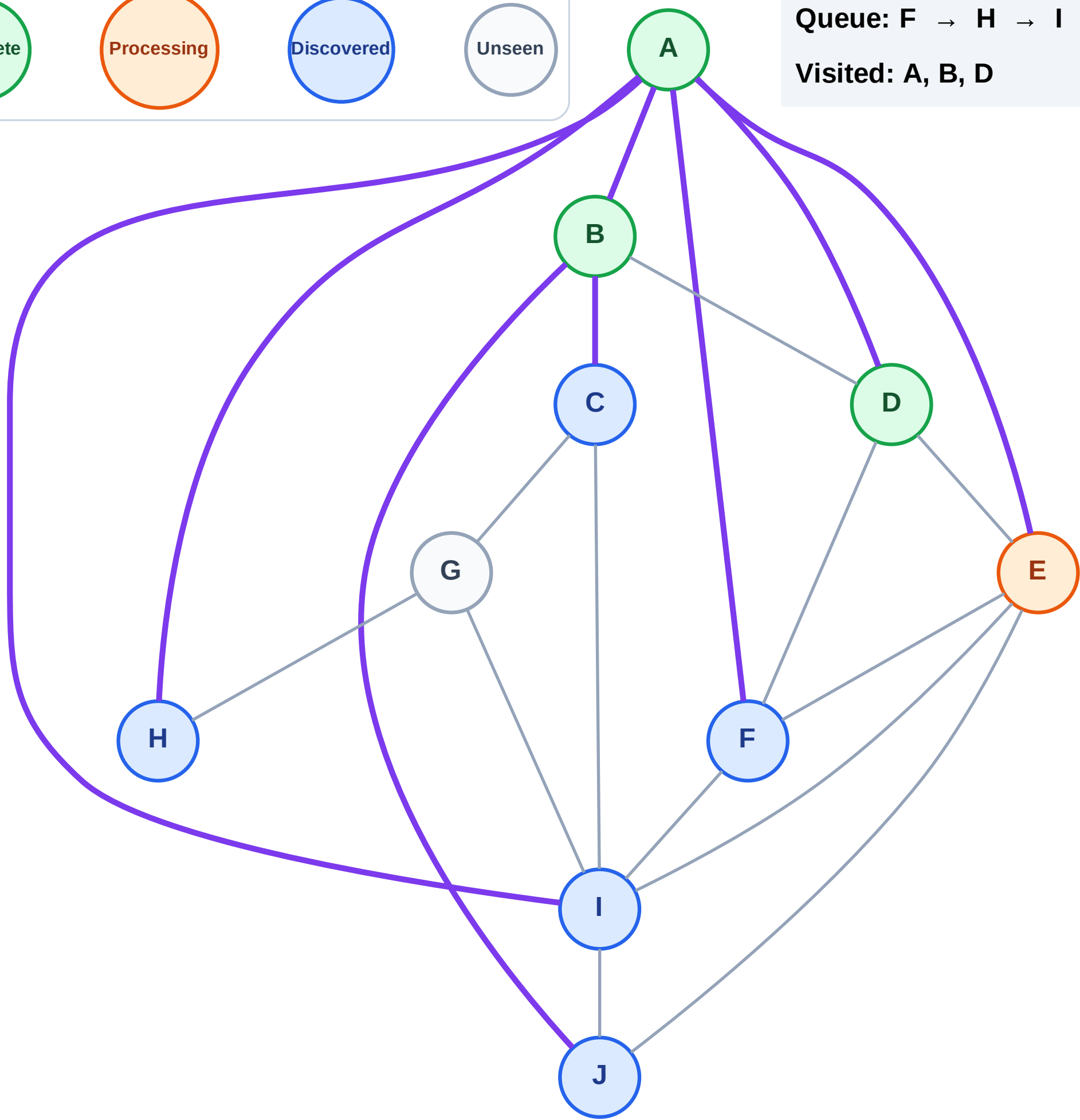
Step 8: Process node E

Legend



Queue: F → H → I → C → J

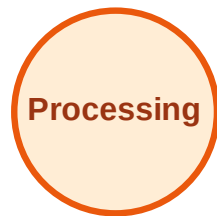
Visited: A, B, D



BFS Traversal

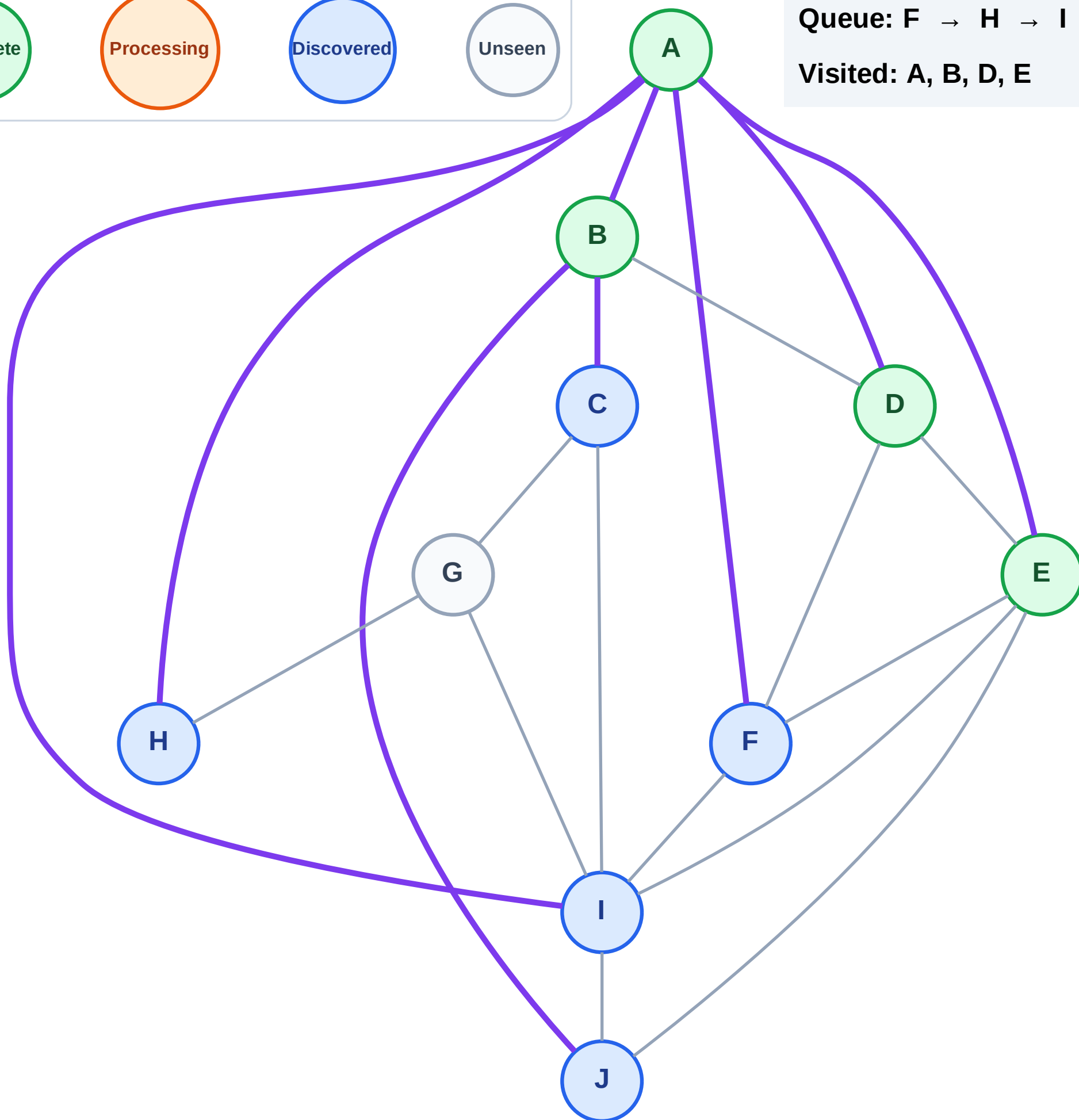
Step 9: No new neighbors from E

Legend



Queue: F → H → I → C → J

Visited: A, B, D, E



BFS Traversal

Step 10: Process node F

Legend

Complete

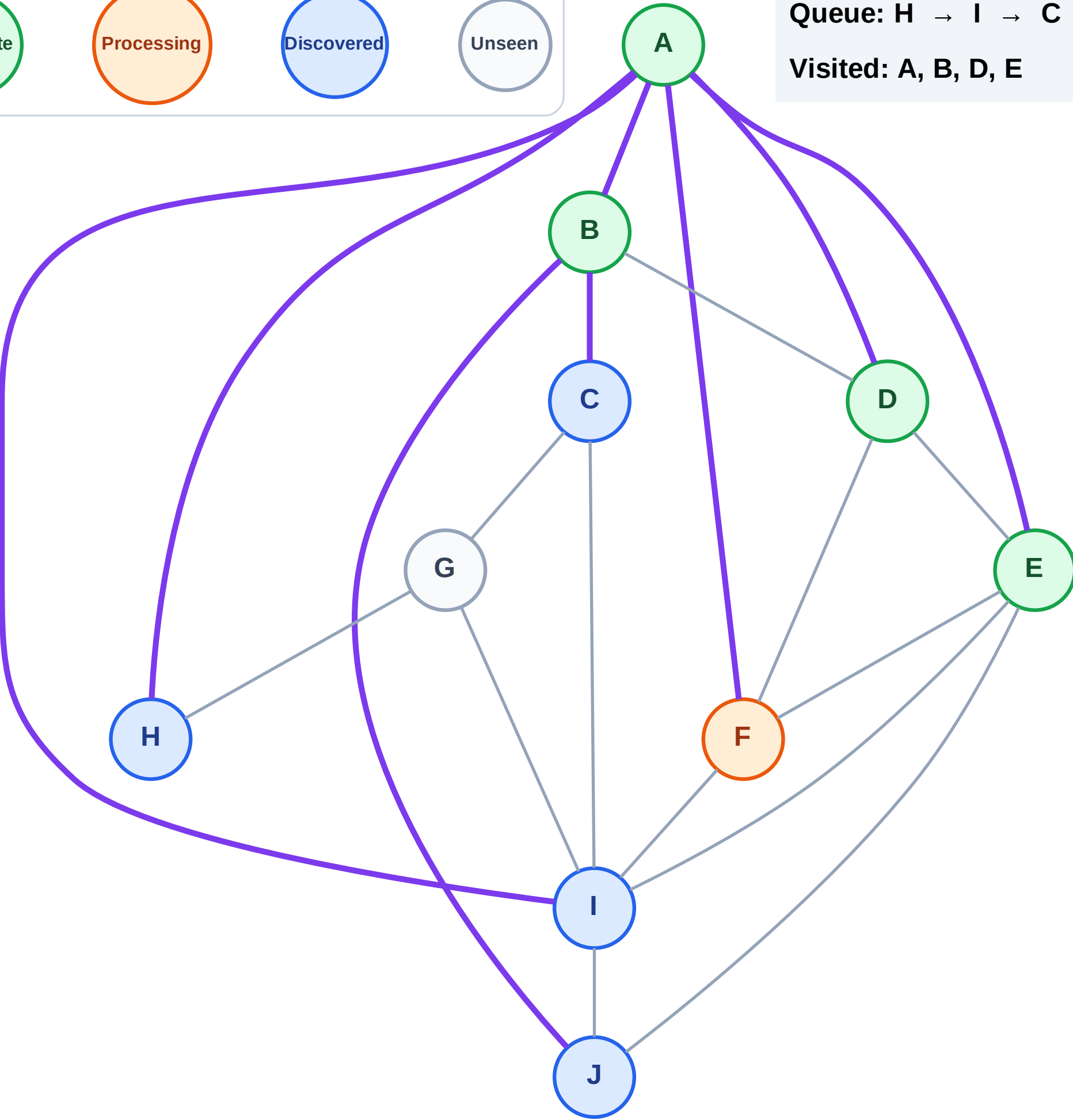
Processing

Discovered

Unseen

Queue: H → I → C → J

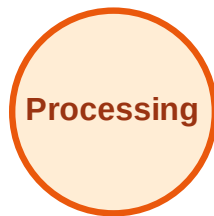
Visited: A, B, D, E



BFS Traversal

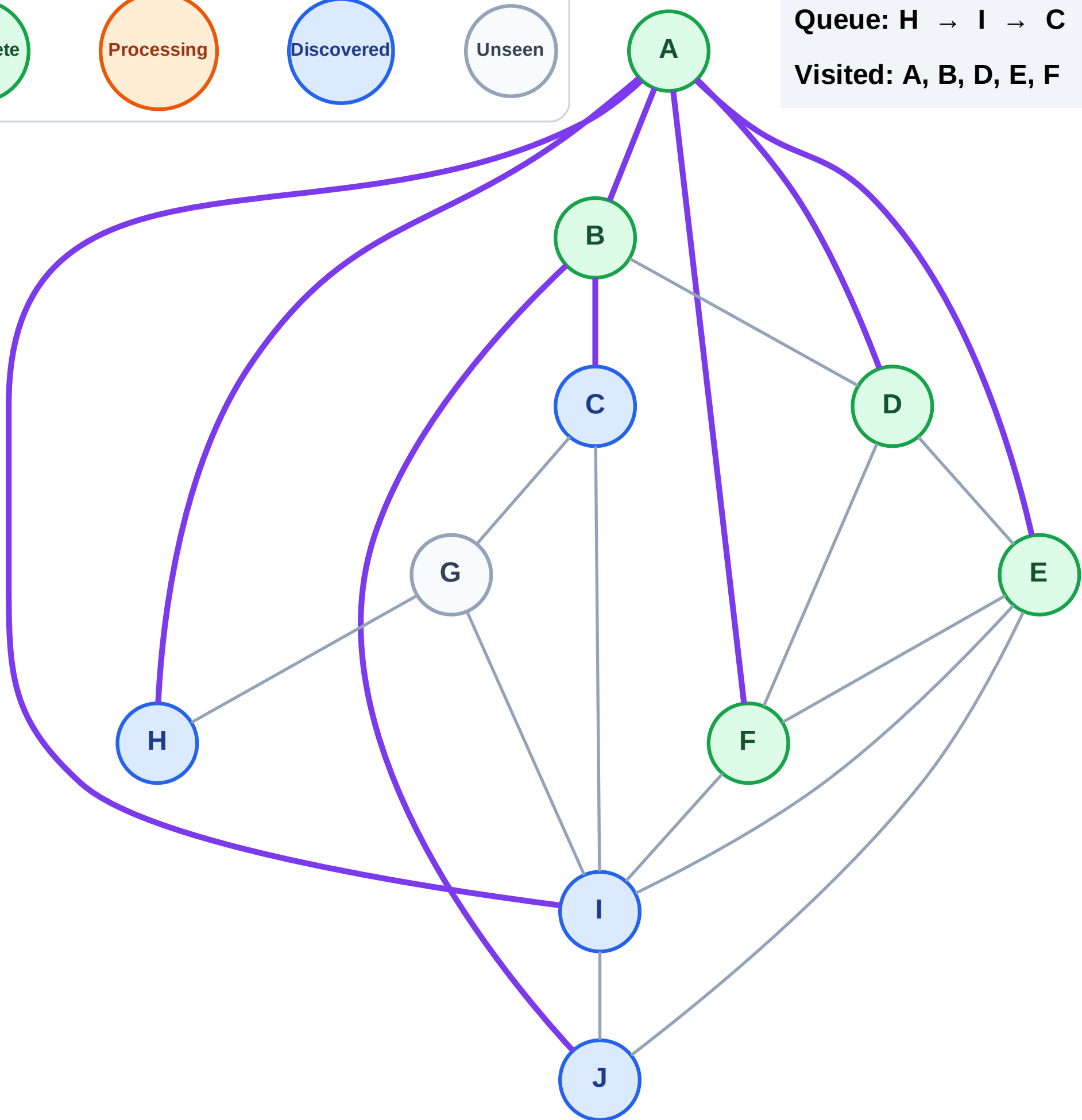
Step 11: No new neighbors from F

Legend



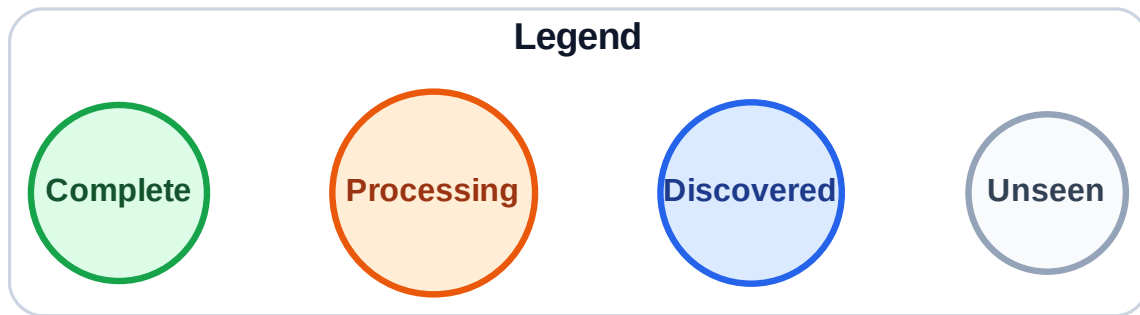
Queue: H → I → C → J

Visited: A, B, D, E, F

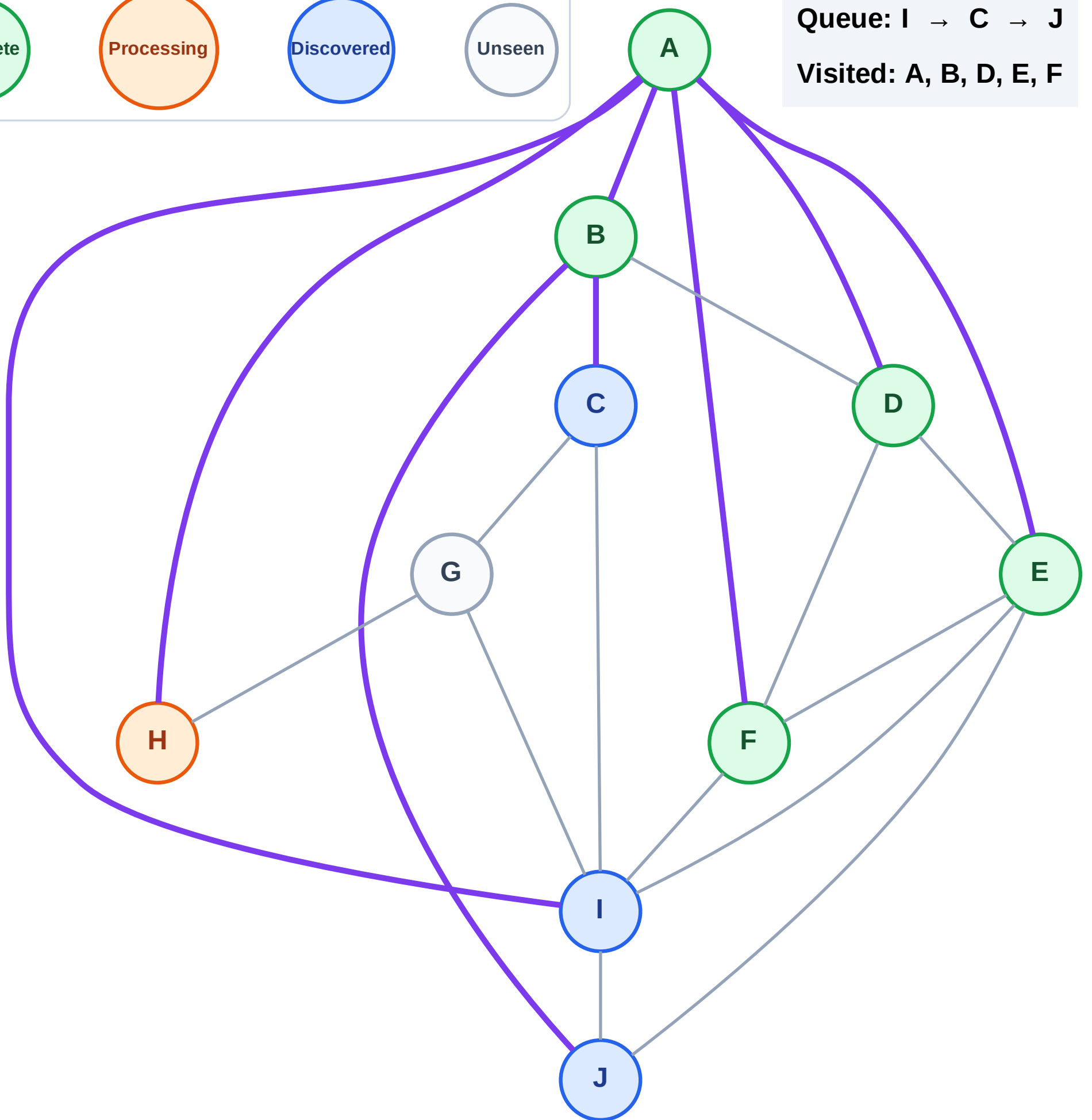


BFS Traversal

Step 12: Process node H



Queue: I → C → J
Visited: A, B, D, E, F

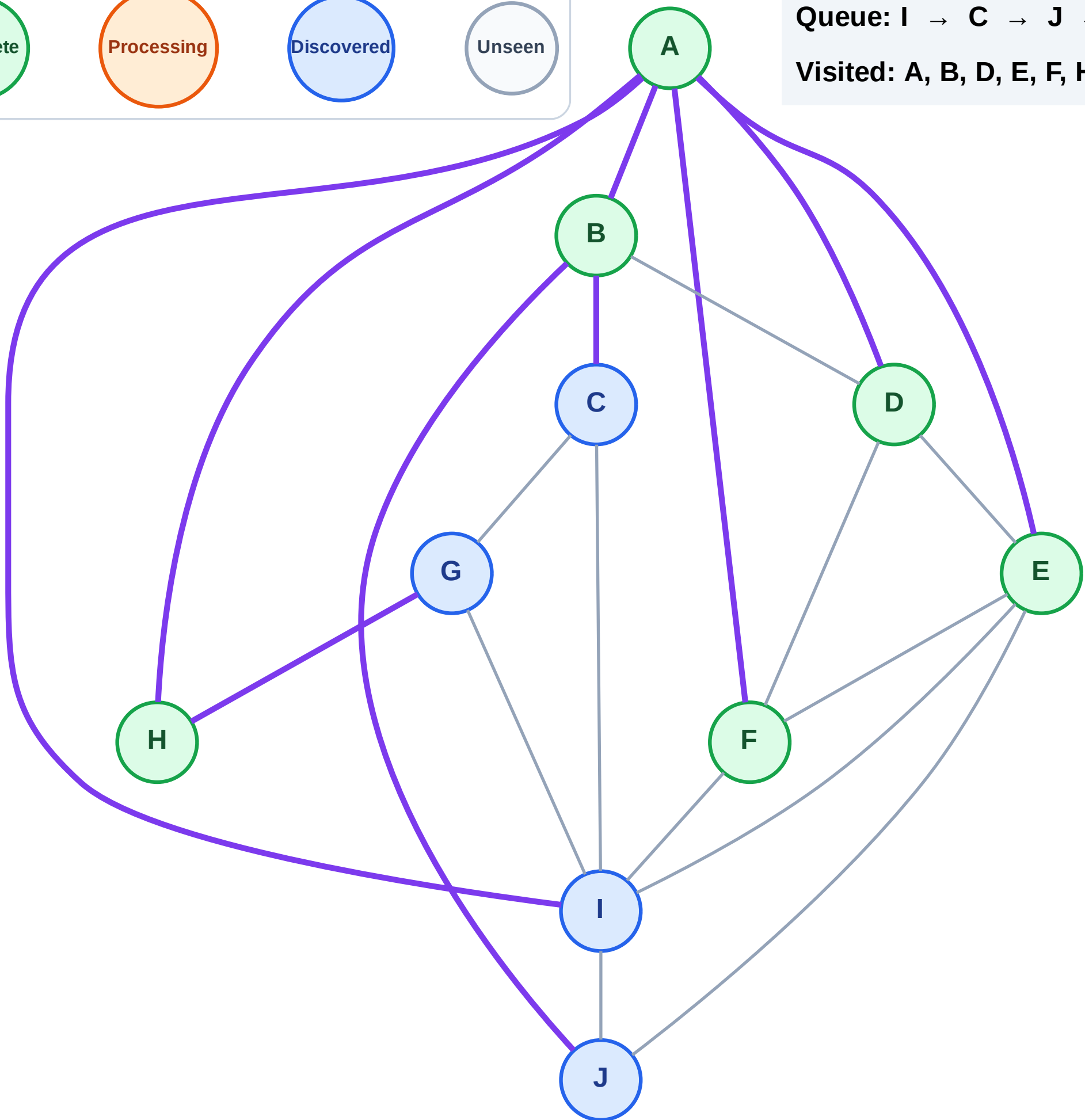


Step 13: Discover G from H

Legend



Visited: A, B, D, E, F, H



BFS Traversal

Step 14: Process node I

Legend

Complete

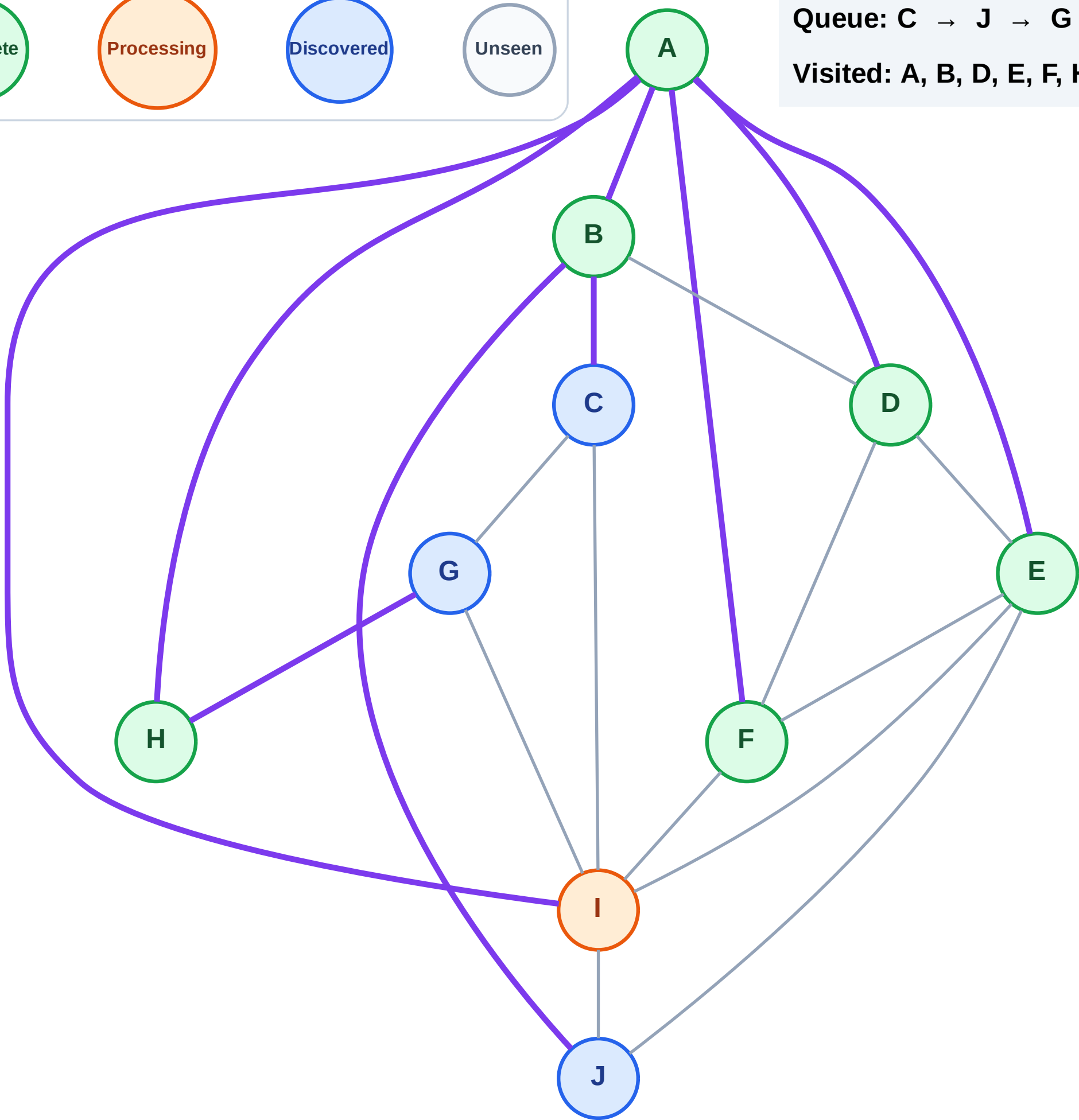
Processing

Discovered

Unseen

Queue: C → J → G

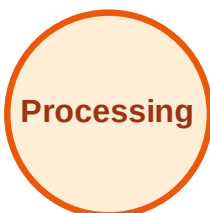
Visited: A, B, D, E, F, H



BFS Traversal

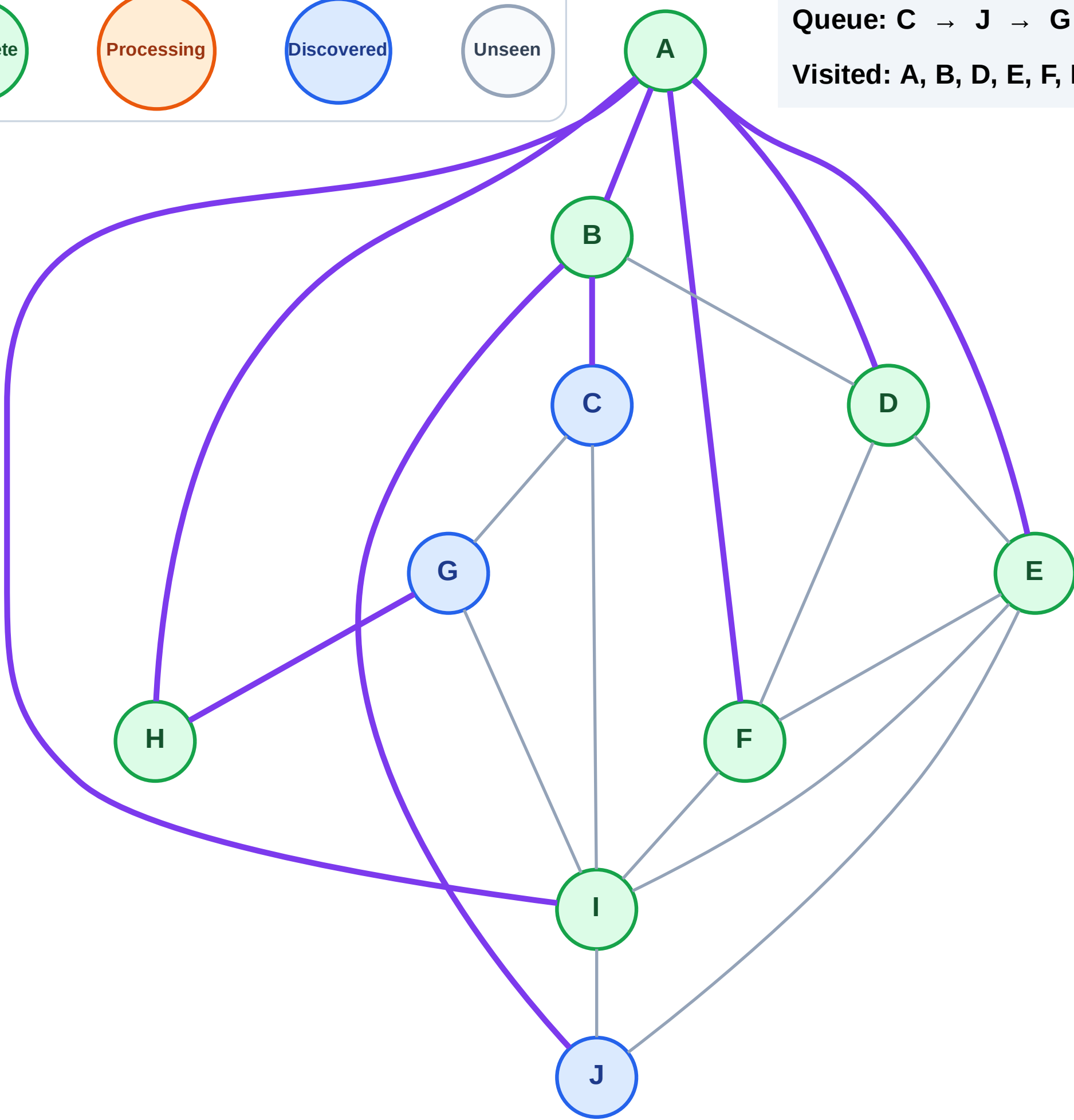
Step 15: No new neighbors from I

Legend

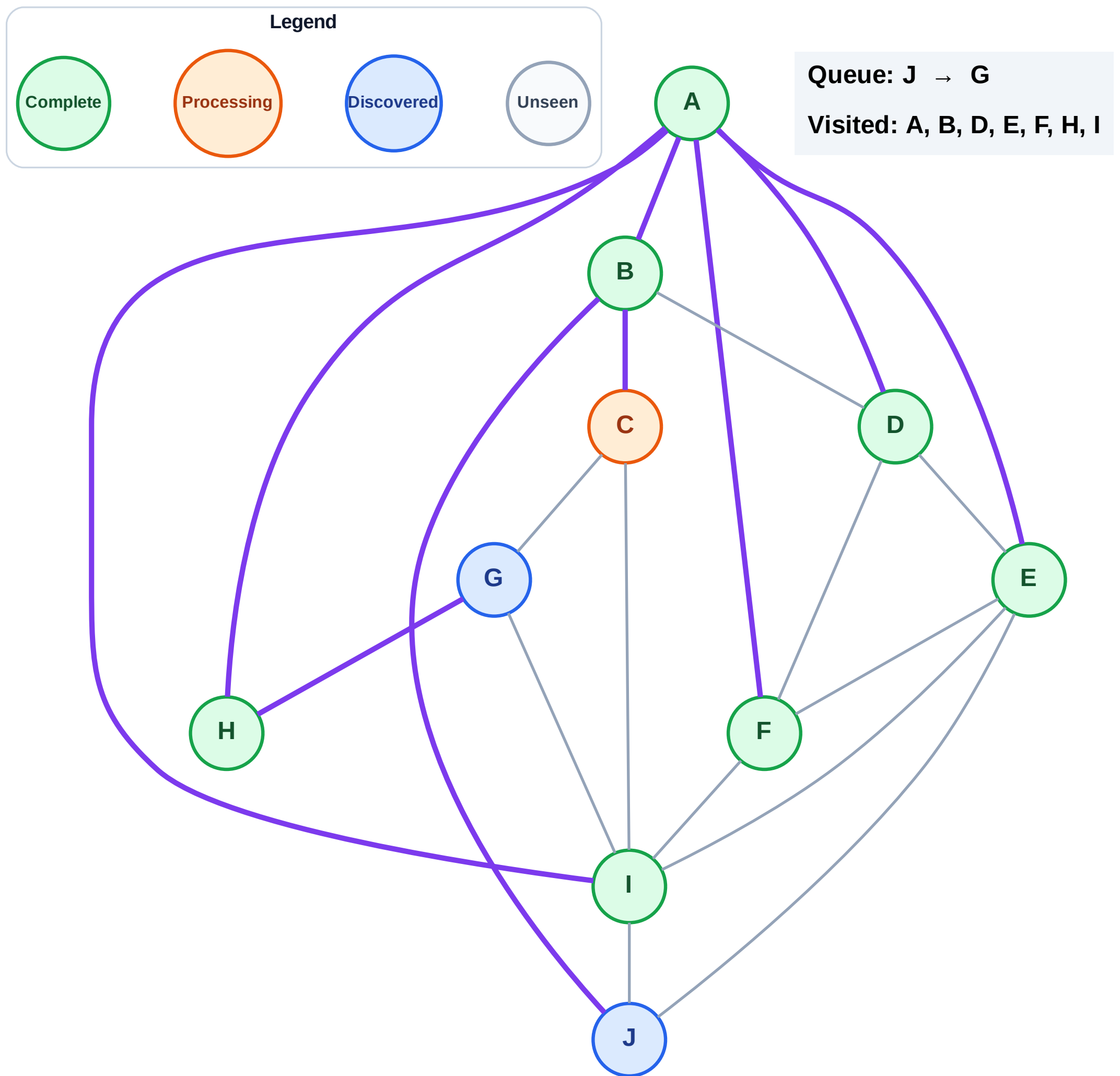


Queue: C → J → G

Visited: A, B, D, E, F, H, I



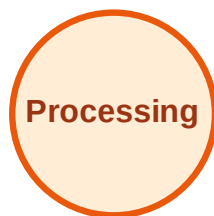
Step 16: Process node C



BFS Traversal

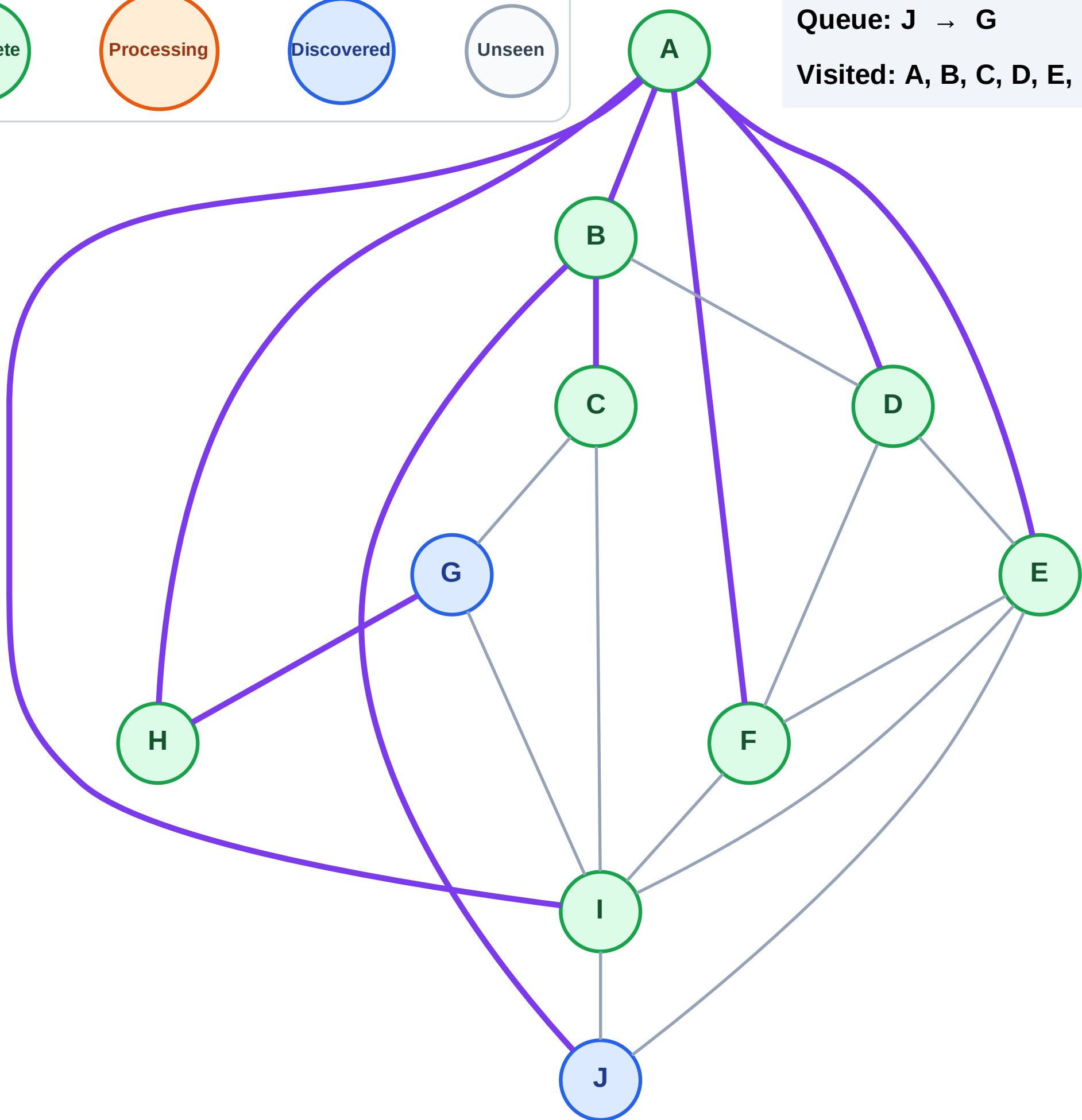
Step 17: No new neighbors from C

Legend



Queue: J → G

Visited: A, B, C, D, E, F, H, I



BFS Traversal

Step 18: Process node J

Legend

Complete

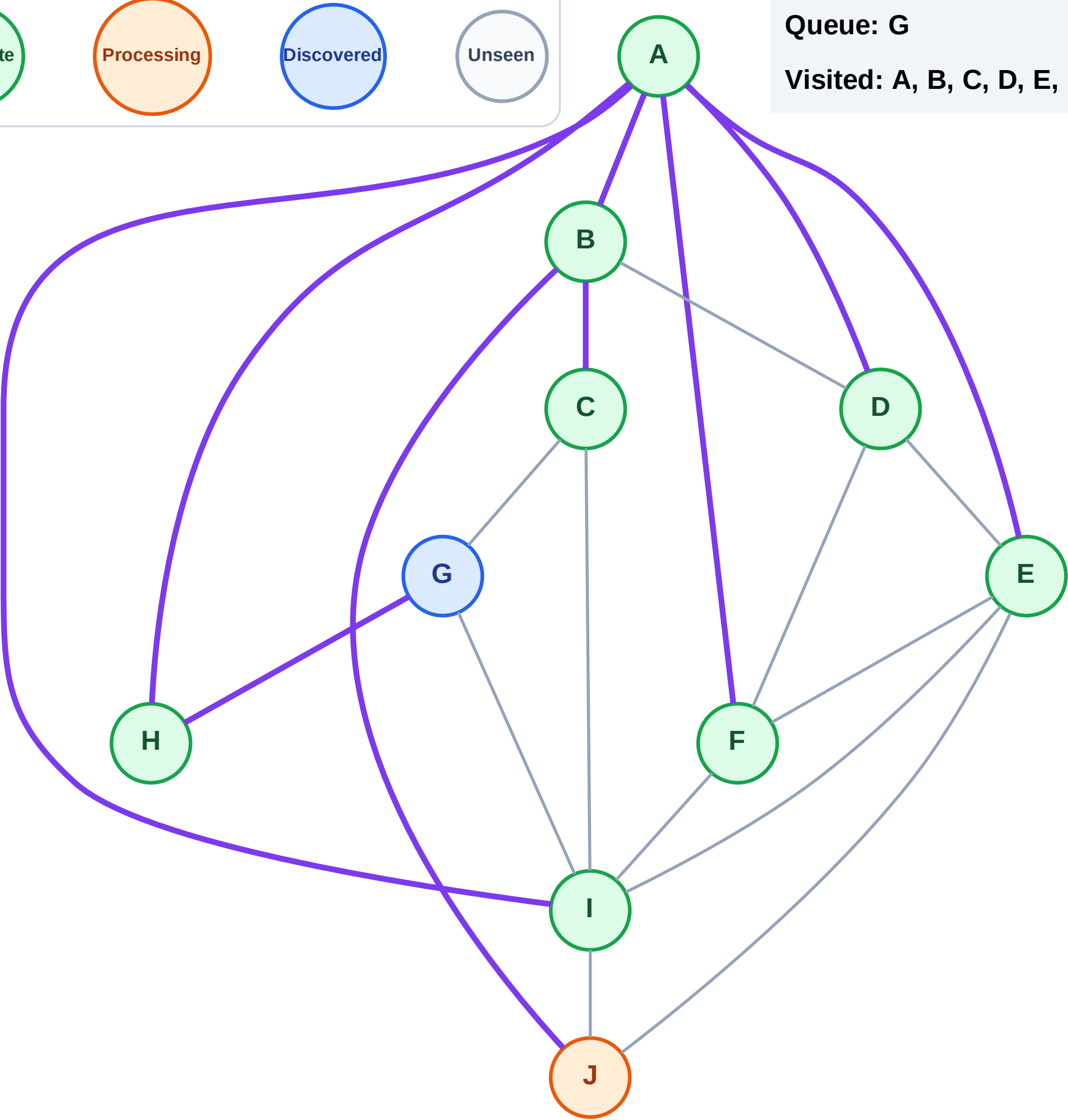
Processing

Discovered

Unseen

Queue: G

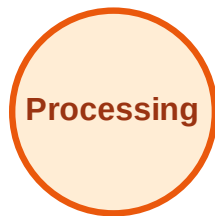
Visited: A, B, C, D, E, F, H, I



BFS Traversal

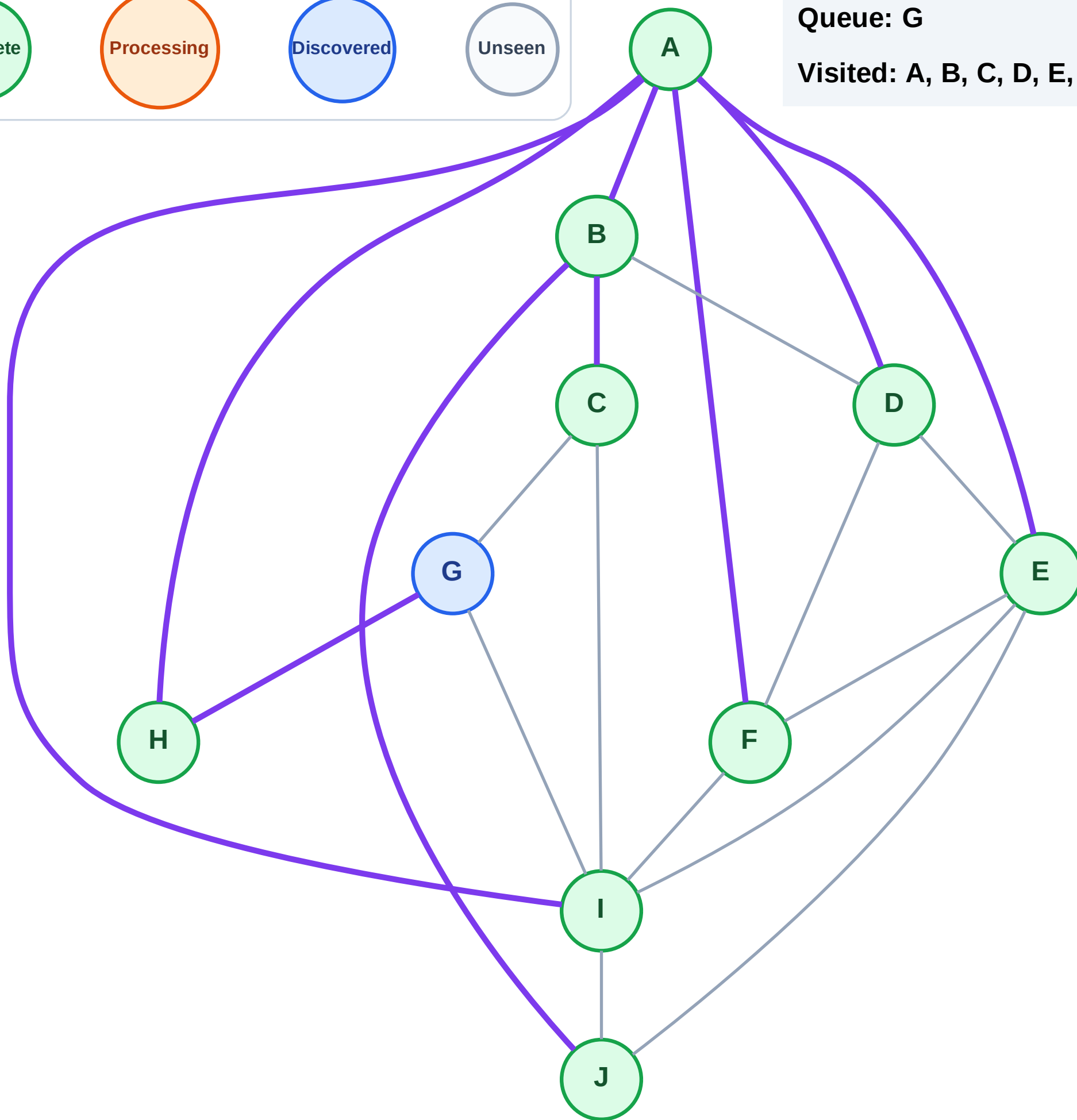
Step 19: No new neighbors from J

Legend



Queue: G

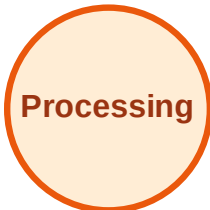
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

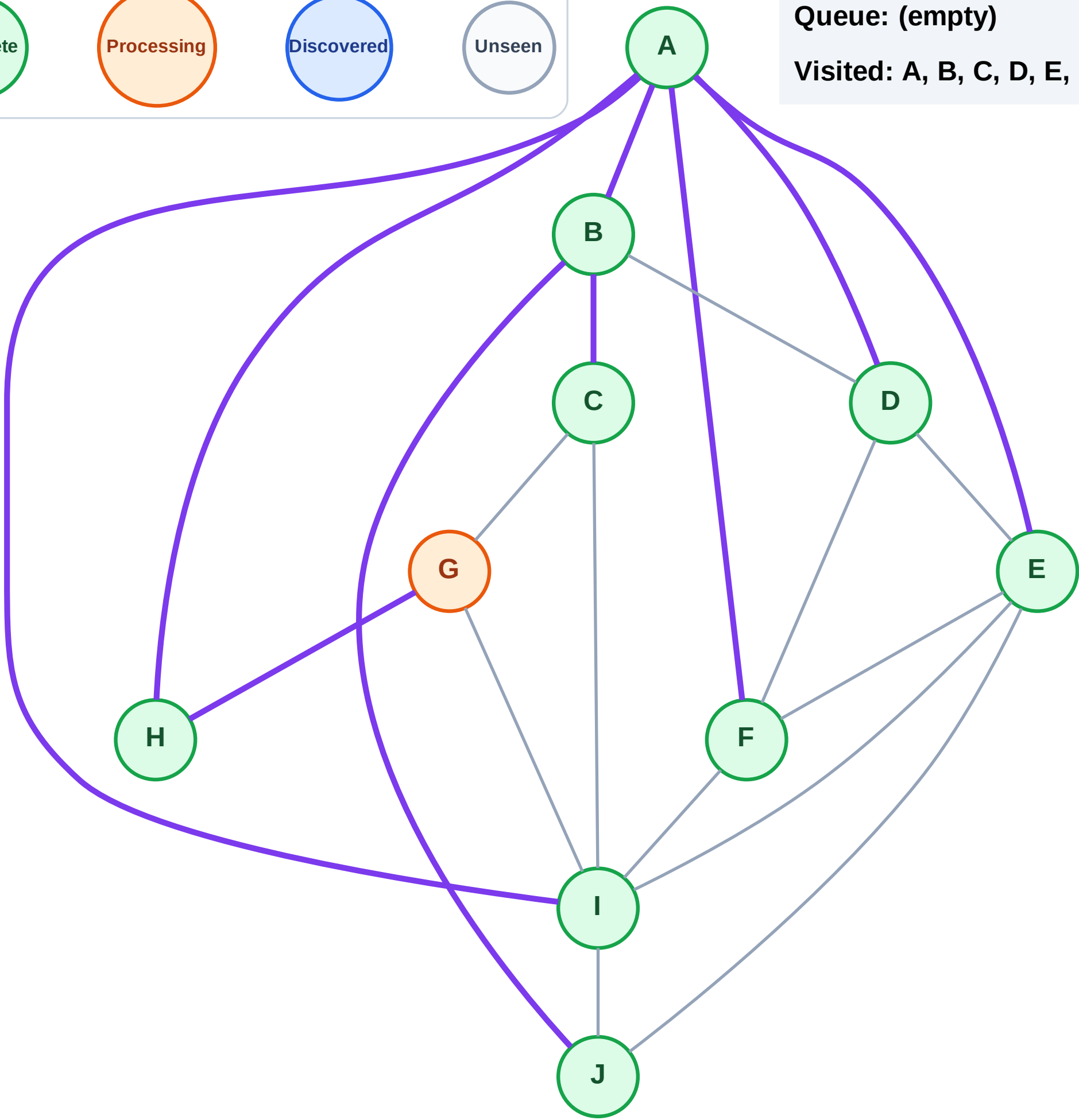
Step 20: Process node G

Legend



Queue: (empty)

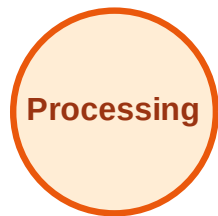
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

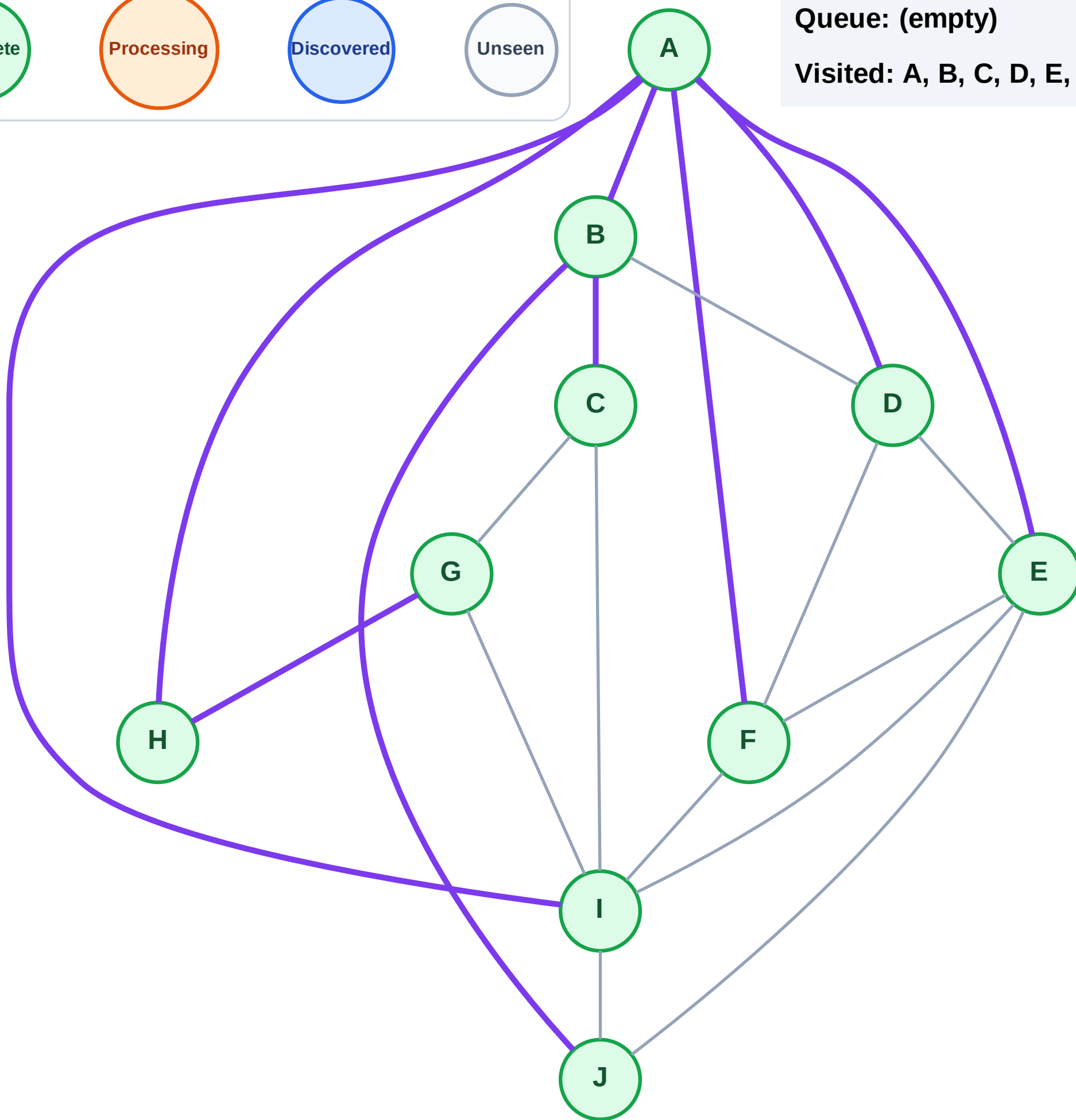
Step 21: No new neighbors from G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

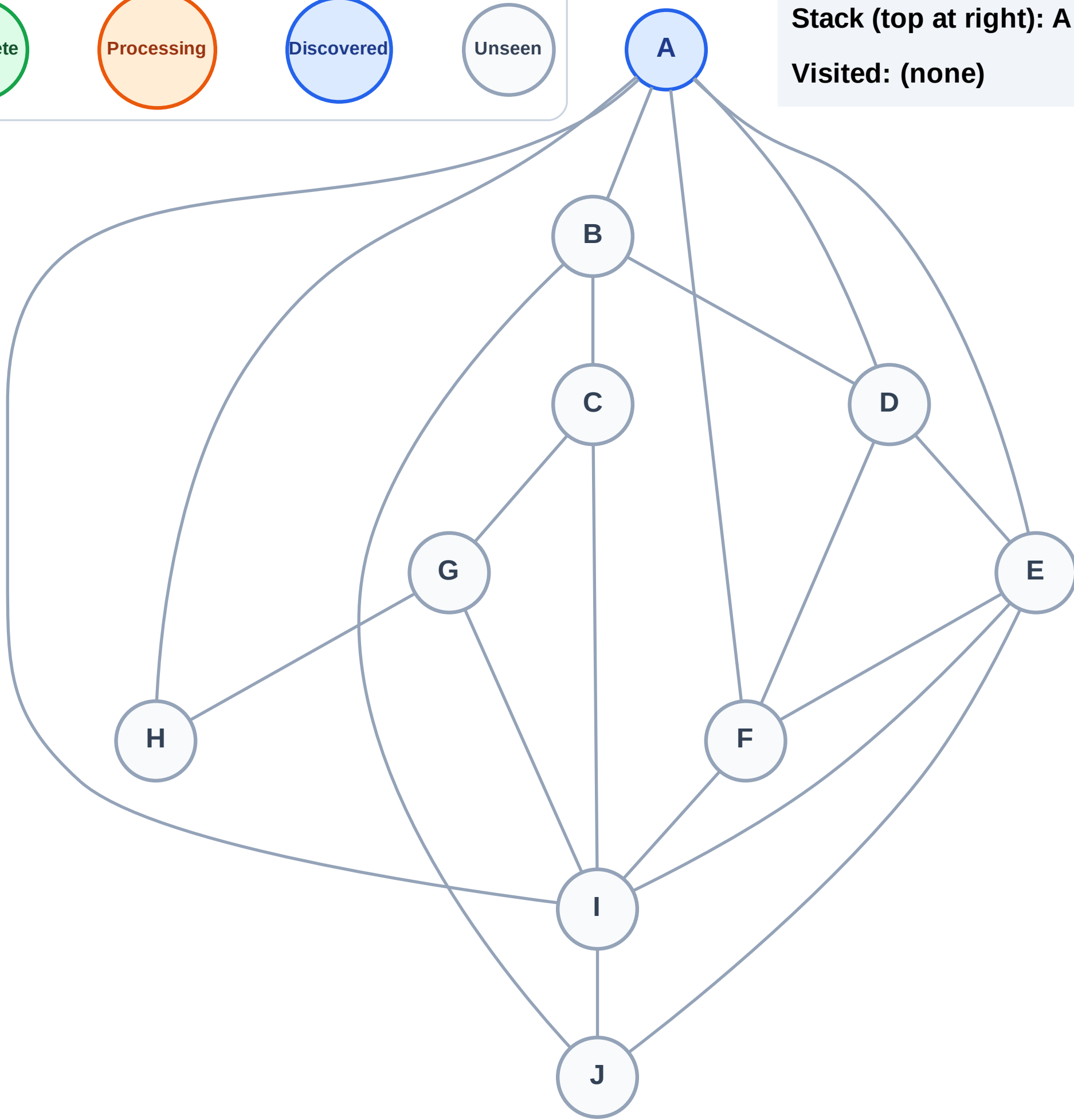
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

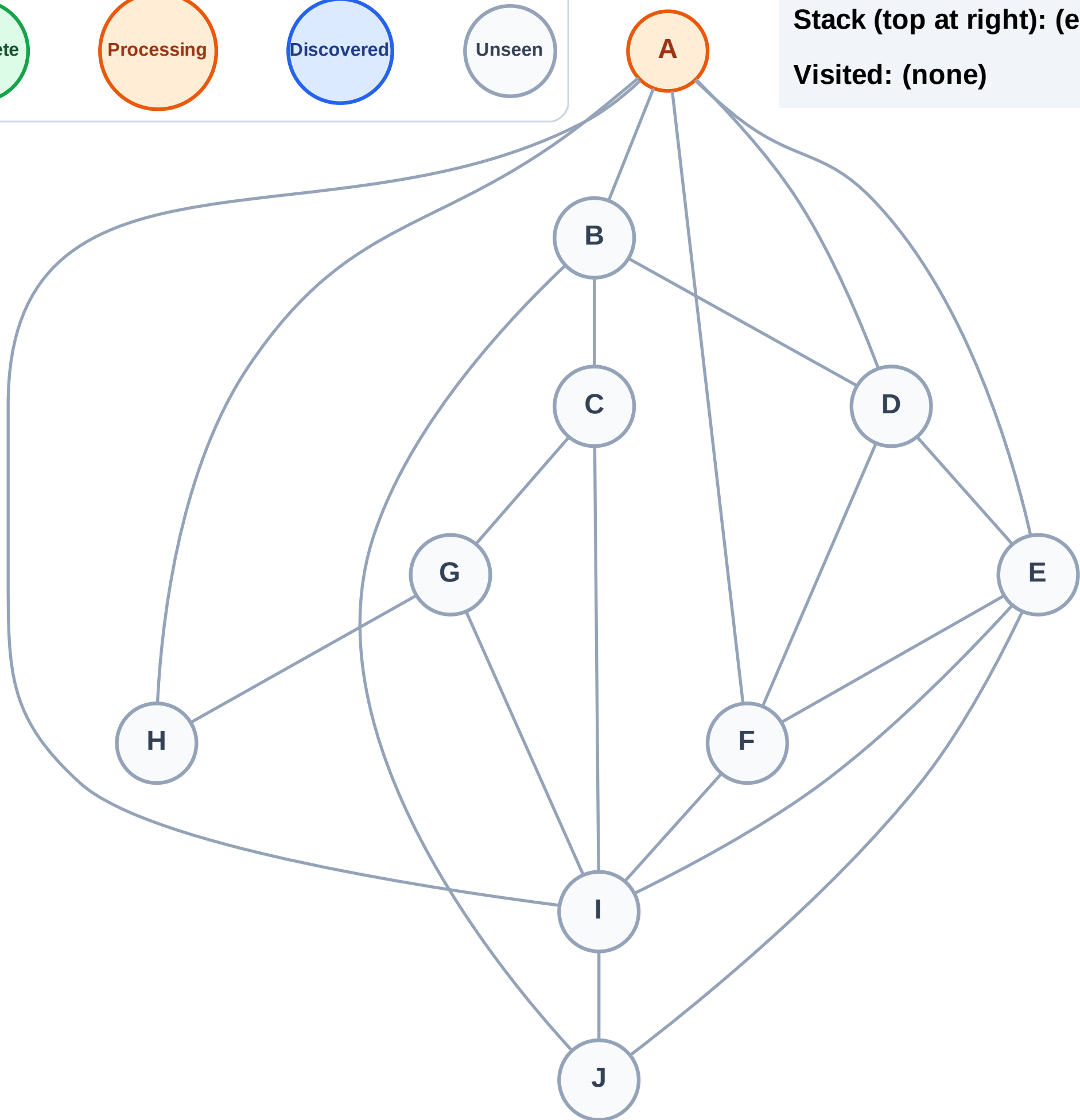
Processing

Discovered

Unseen

Stack (top at right): (empty)

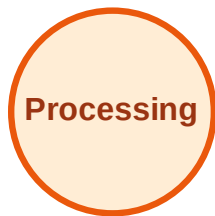
Visited: (none)



DFS Traversal

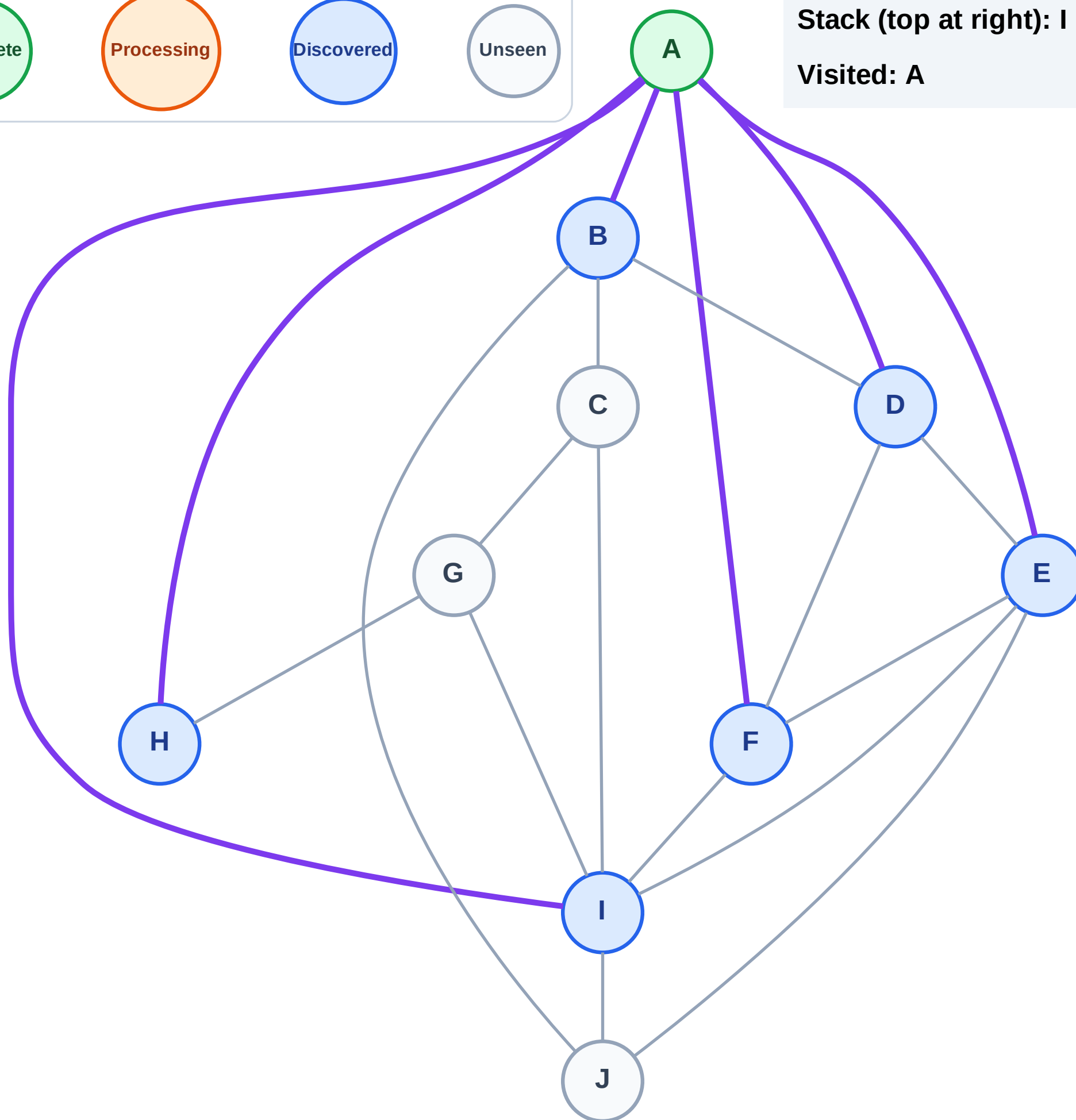
Step 3: Discover I, H, F, E, D, B from A

Legend



Stack (top at right): I | H | F | E | D | B

Visited: A

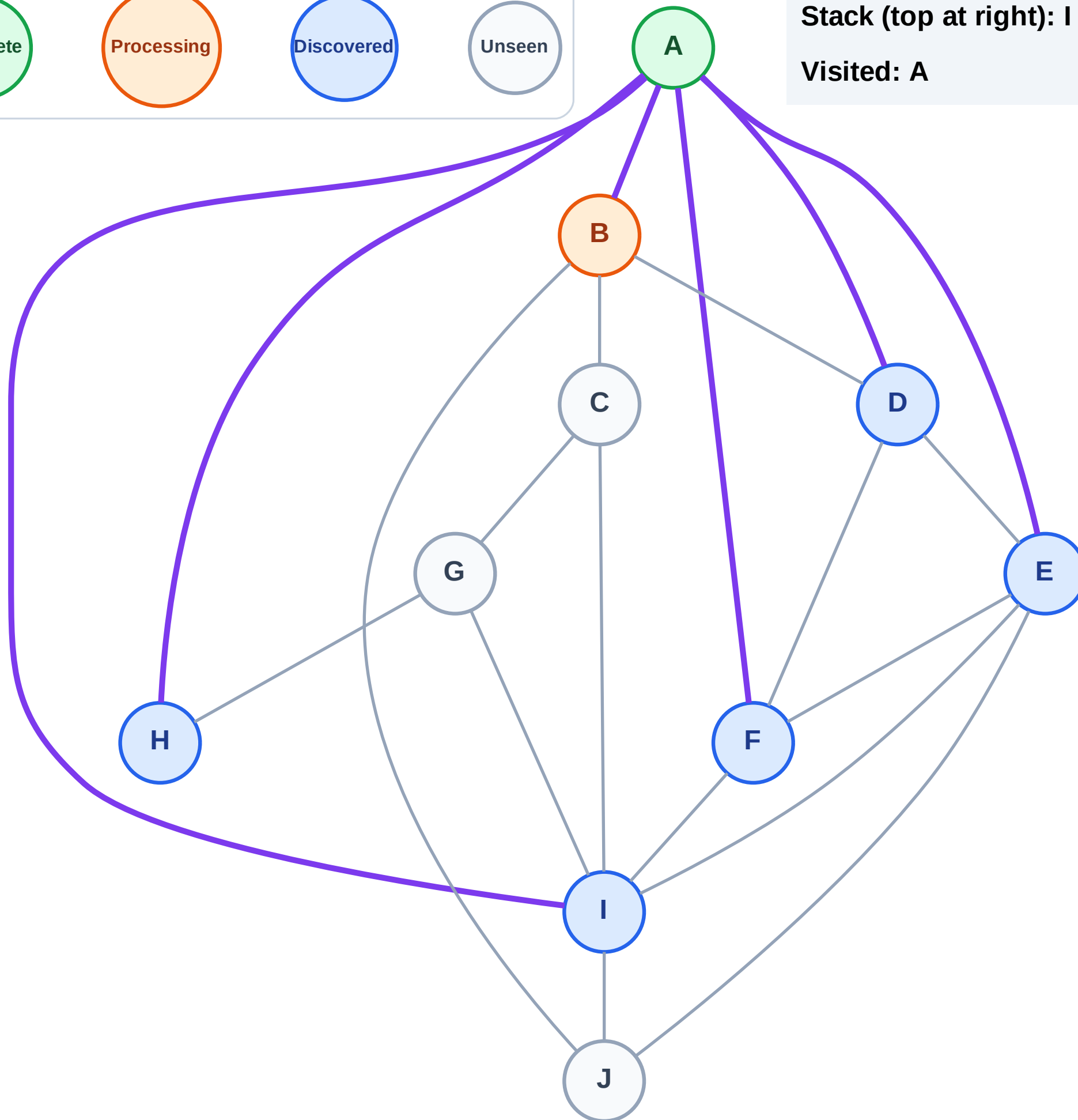
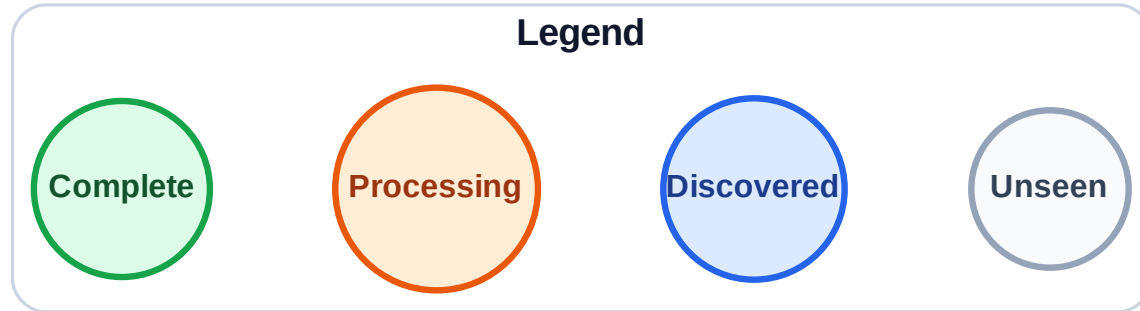


DFS Traversal

Step 4: Process node B

Stack (top at right): I | H | F | E | D

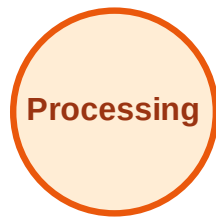
Visited: A



DFS Traversal

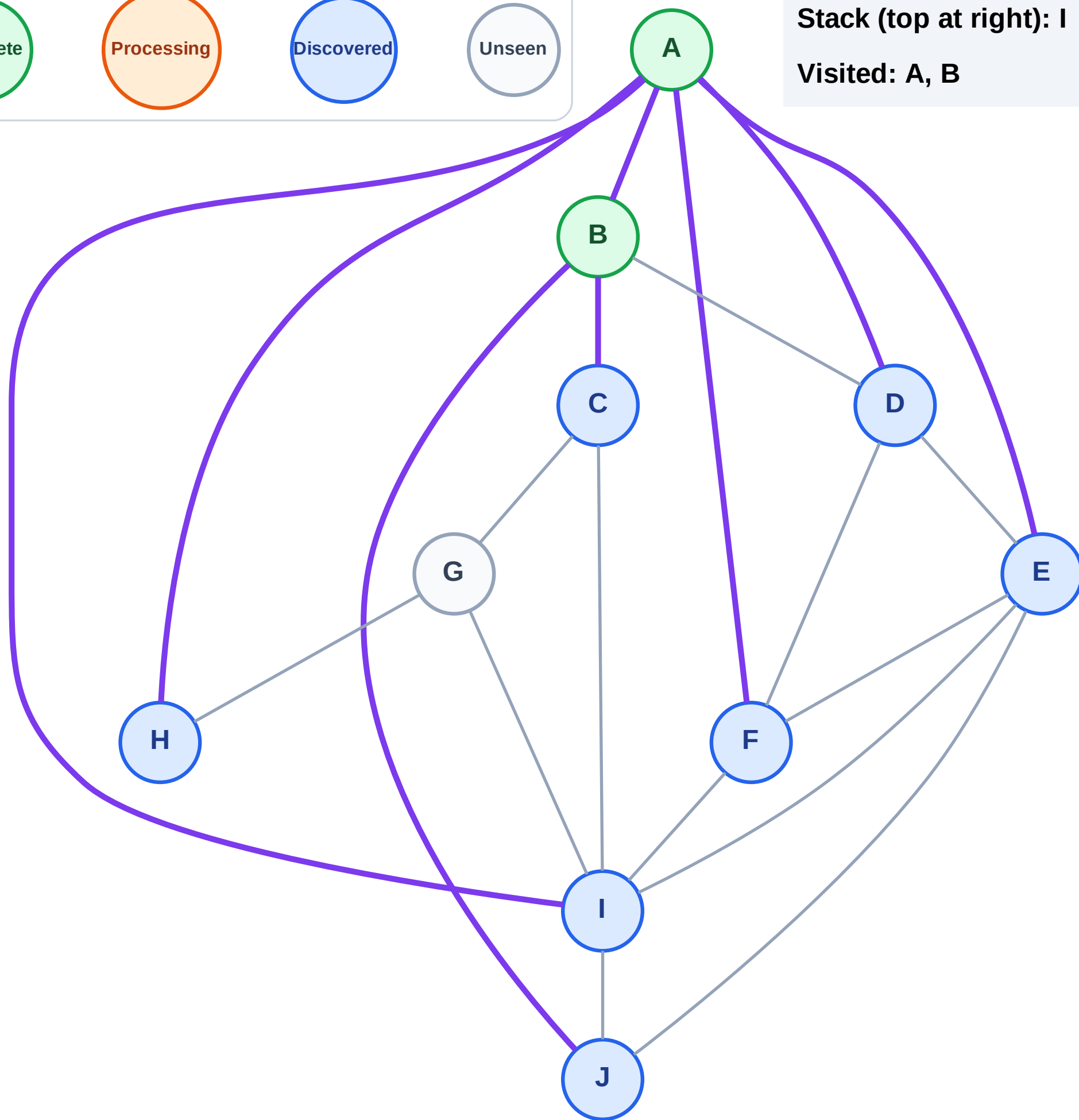
Step 5: Discover J, C from B

Legend



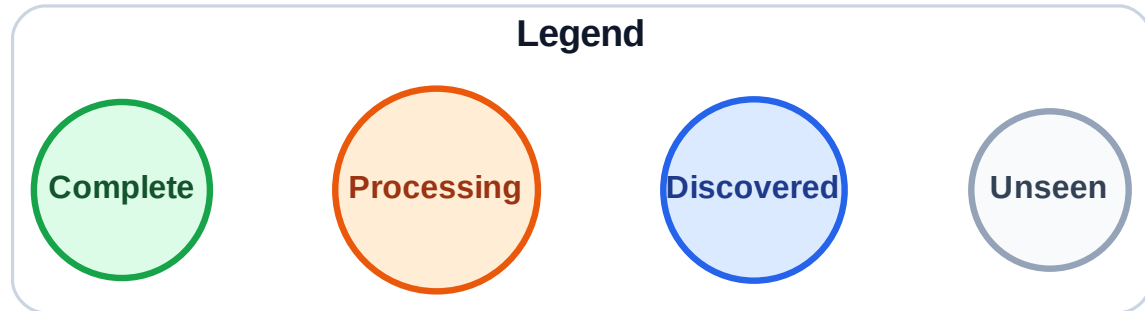
Stack (top at right): I | H | F | E | D | J | C

Visited: A, B



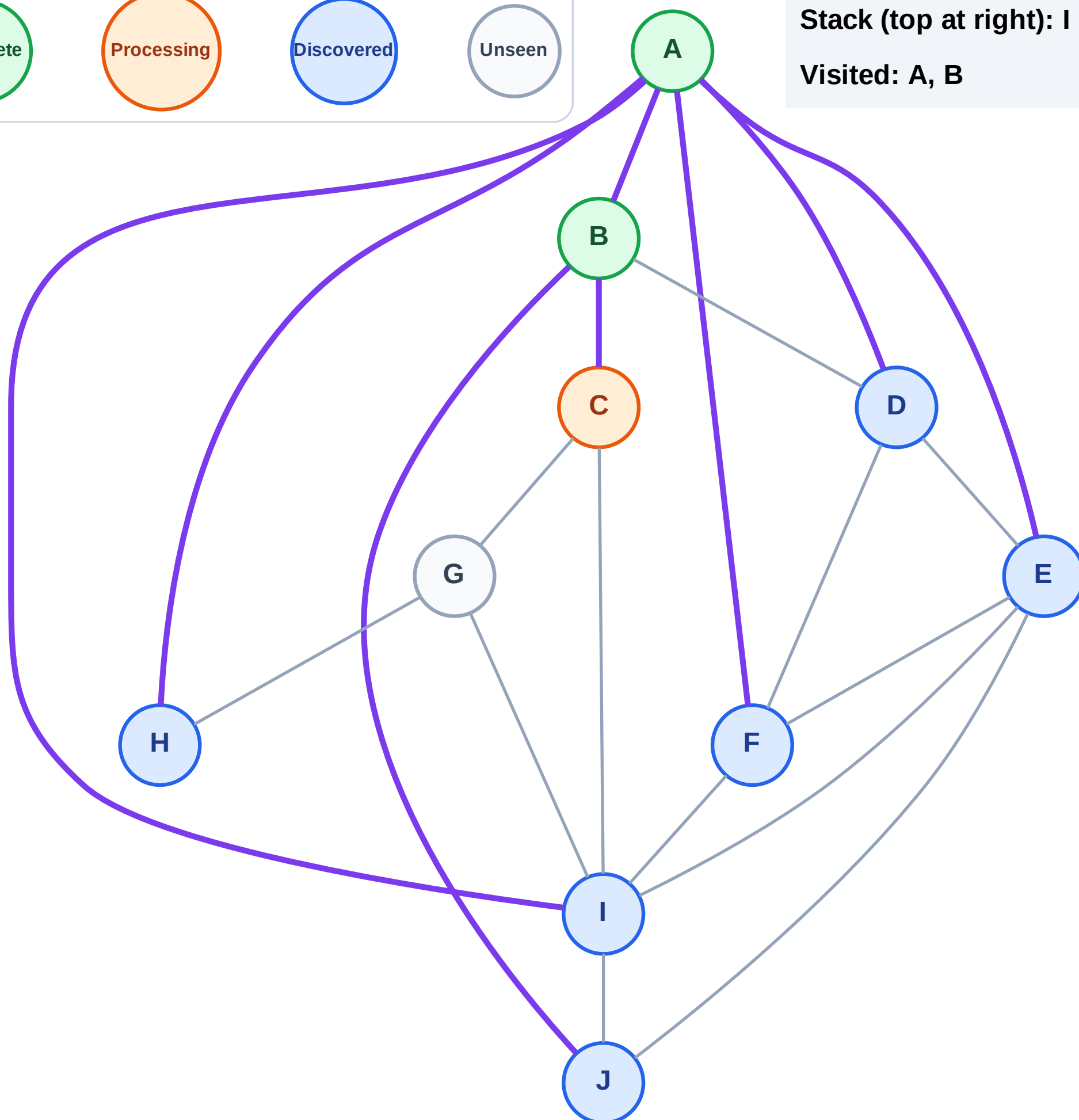
DFS Traversal

Step 6: Process node C



Stack (top at right): I | H | F | E | D | J

Visited: A, B



DFS Traversal

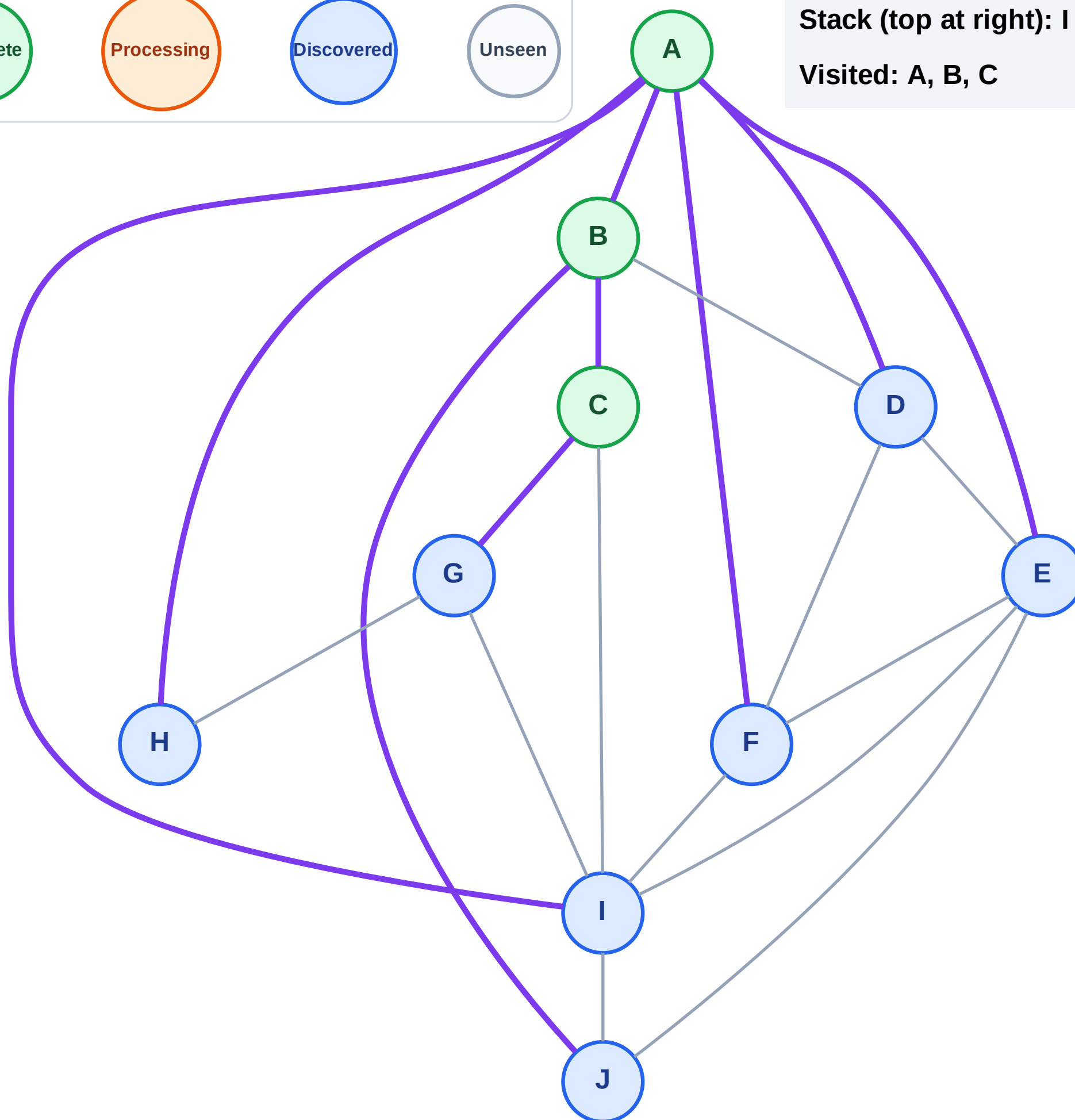
Step 7: Discover G from C

Legend



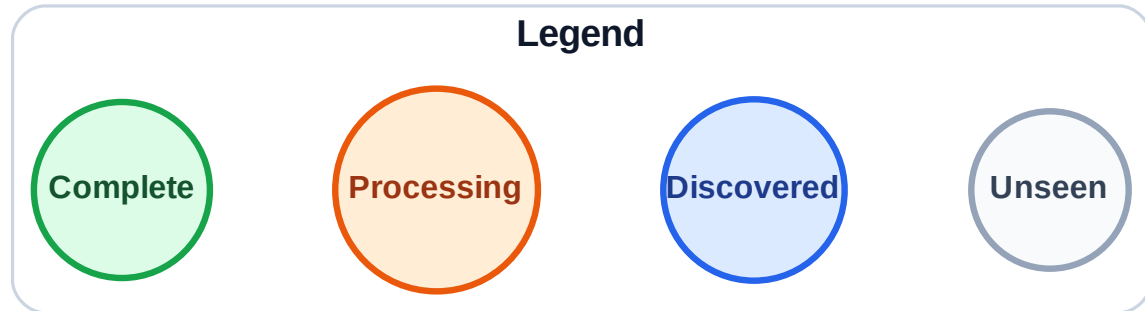
Stack (top at right): I | H | F | E | D | J | G

Visited: A, B, C



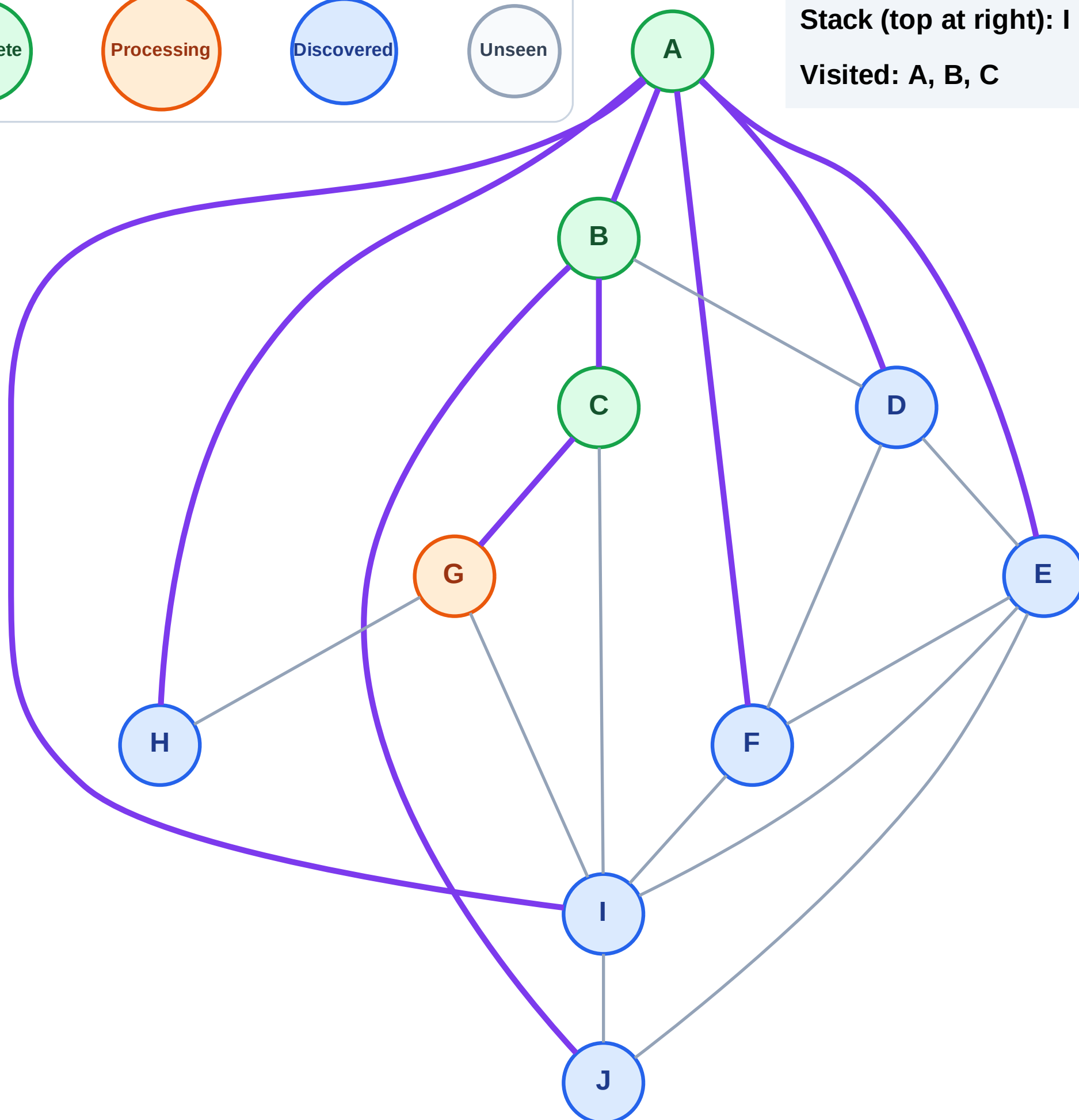
DFS Traversal

Step 8: Process node G



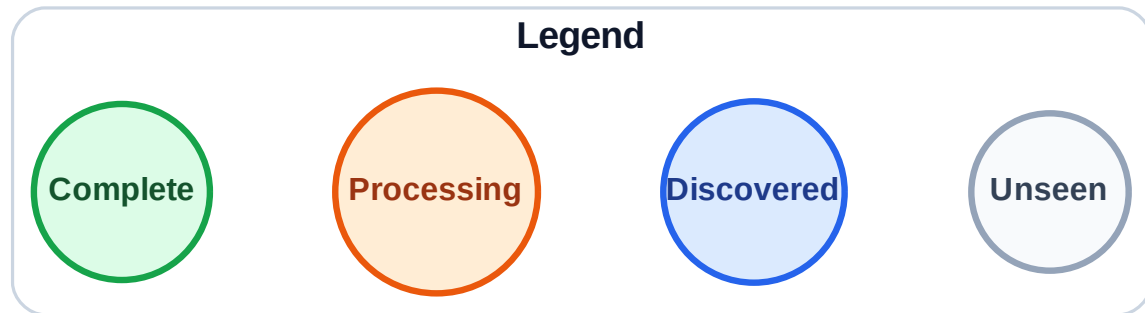
Stack (top at right): I | H | F | E | D | J

Visited: A, B, C



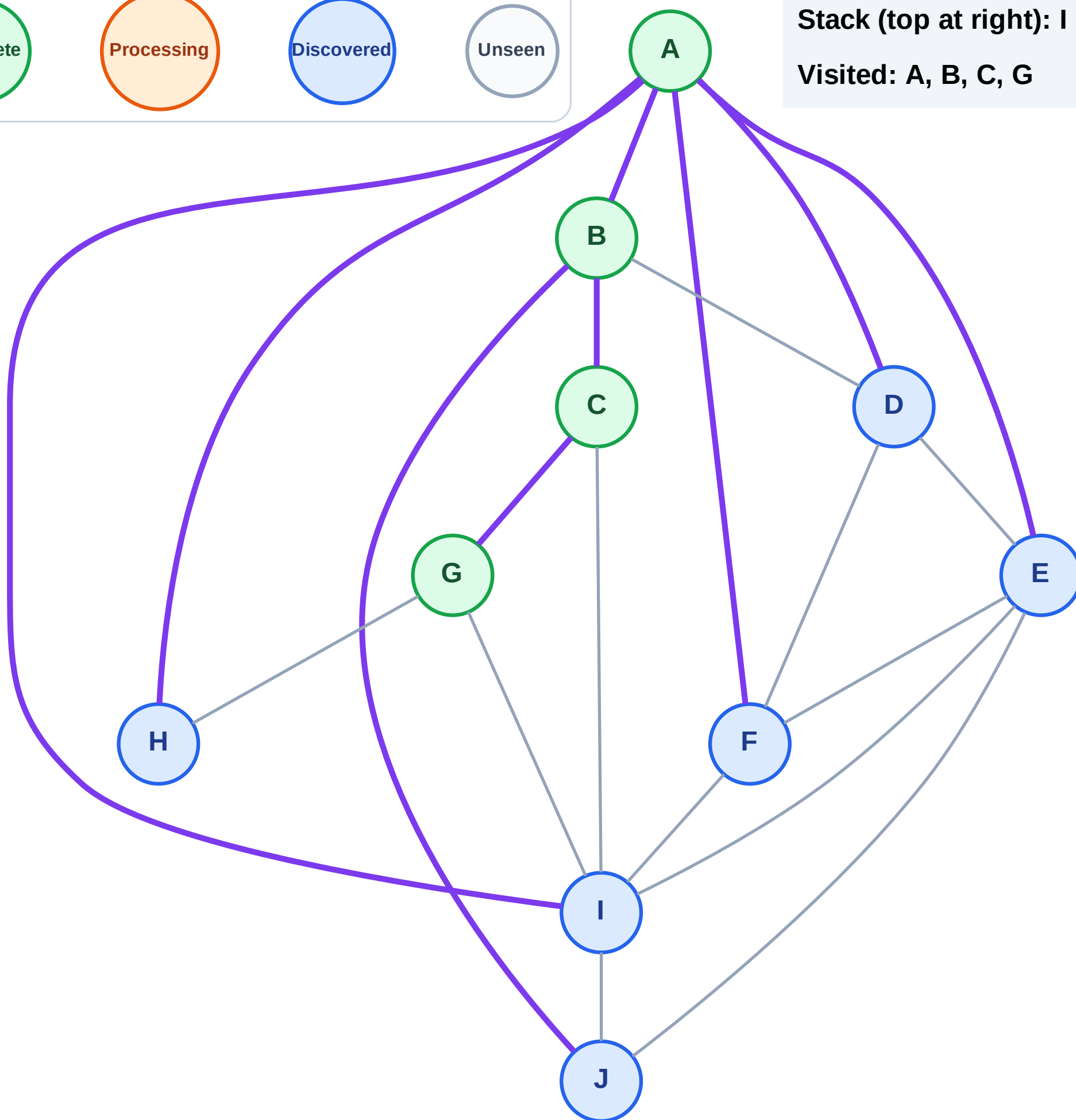
Step 9: No new neighbors from G

Step 9: No new neighbors from G



Stack (top at right): I | H | F | E | D | J

Visited: A, B, C, G



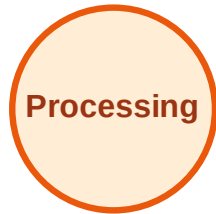
DFS Traversal

Step 10: Process node J

Legend



Complete



Processing



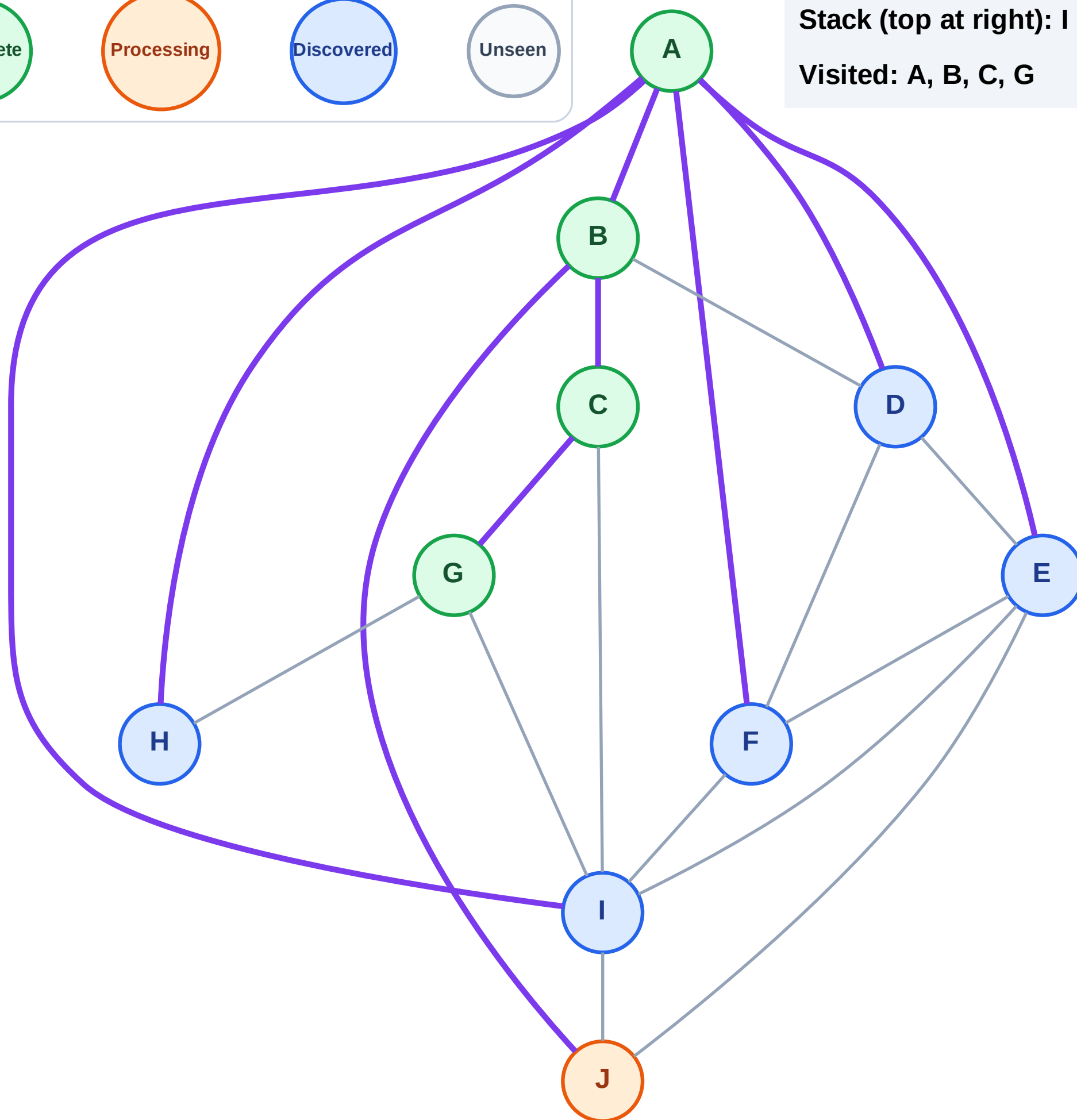
Discovered



Unseen

Stack (top at right): I | H | F | E | D

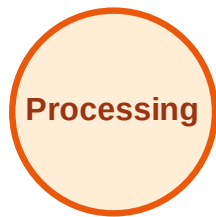
Visited: A, B, C, G



DFS Traversal

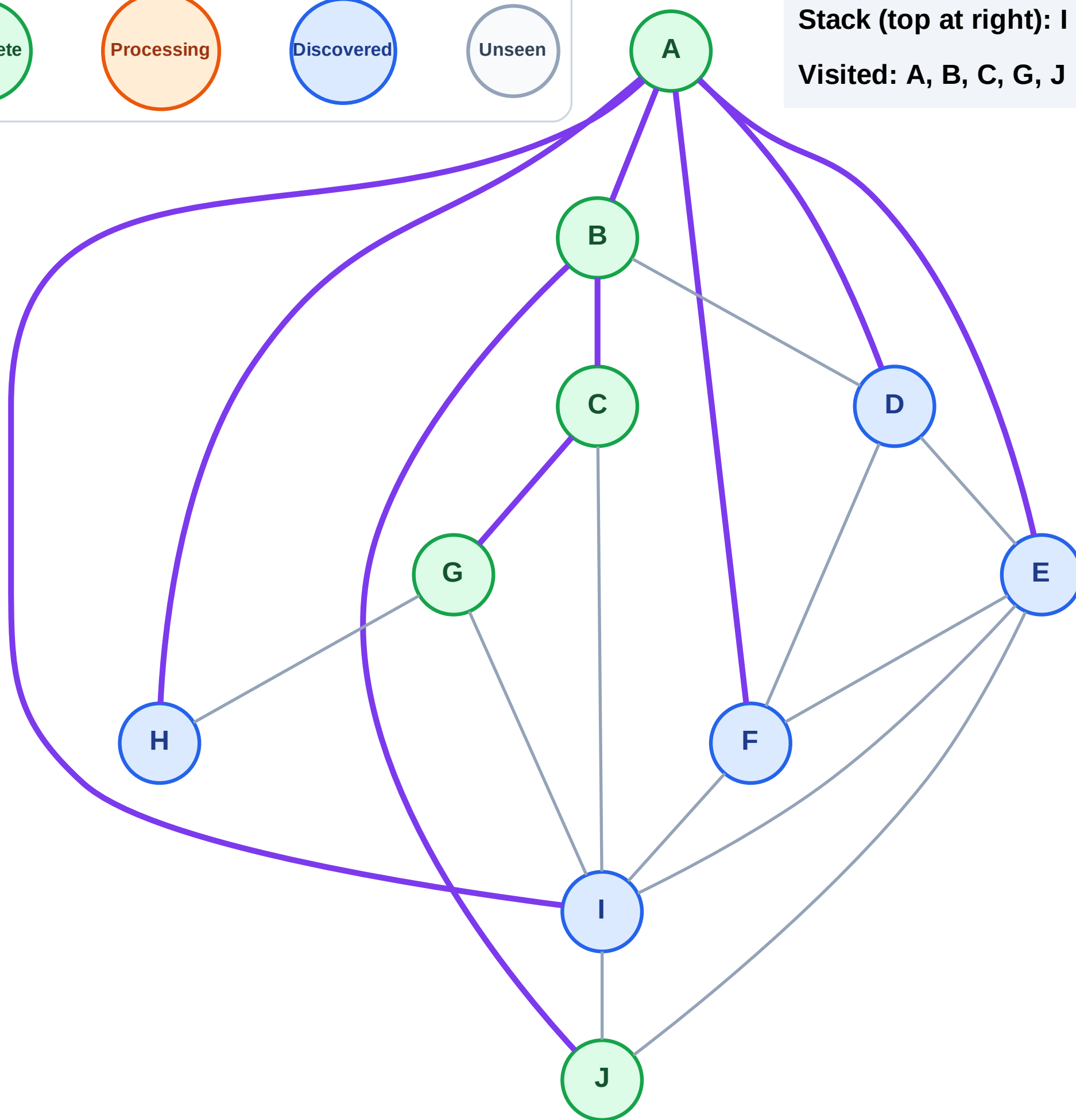
Step 11: No new neighbors from J

Legend



Stack (top at right): I | H | F | E | D

Visited: A, B, C, G, J



DFS Traversal

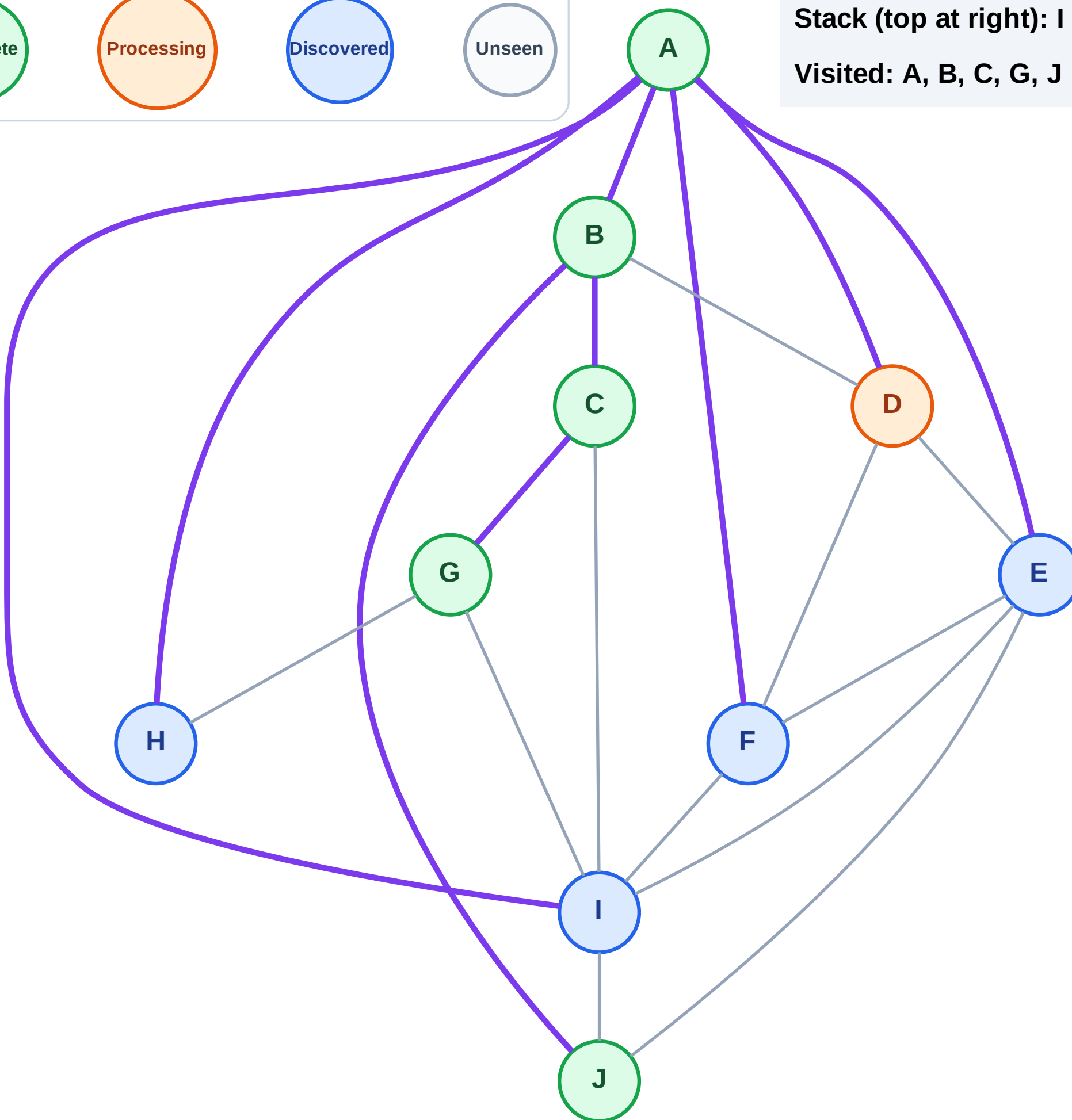
Step 12: Process node D

Legend



Stack (top at right): I | H | F | E

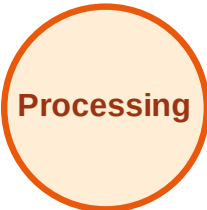
Visited: A, B, C, G, J



DFS Traversal

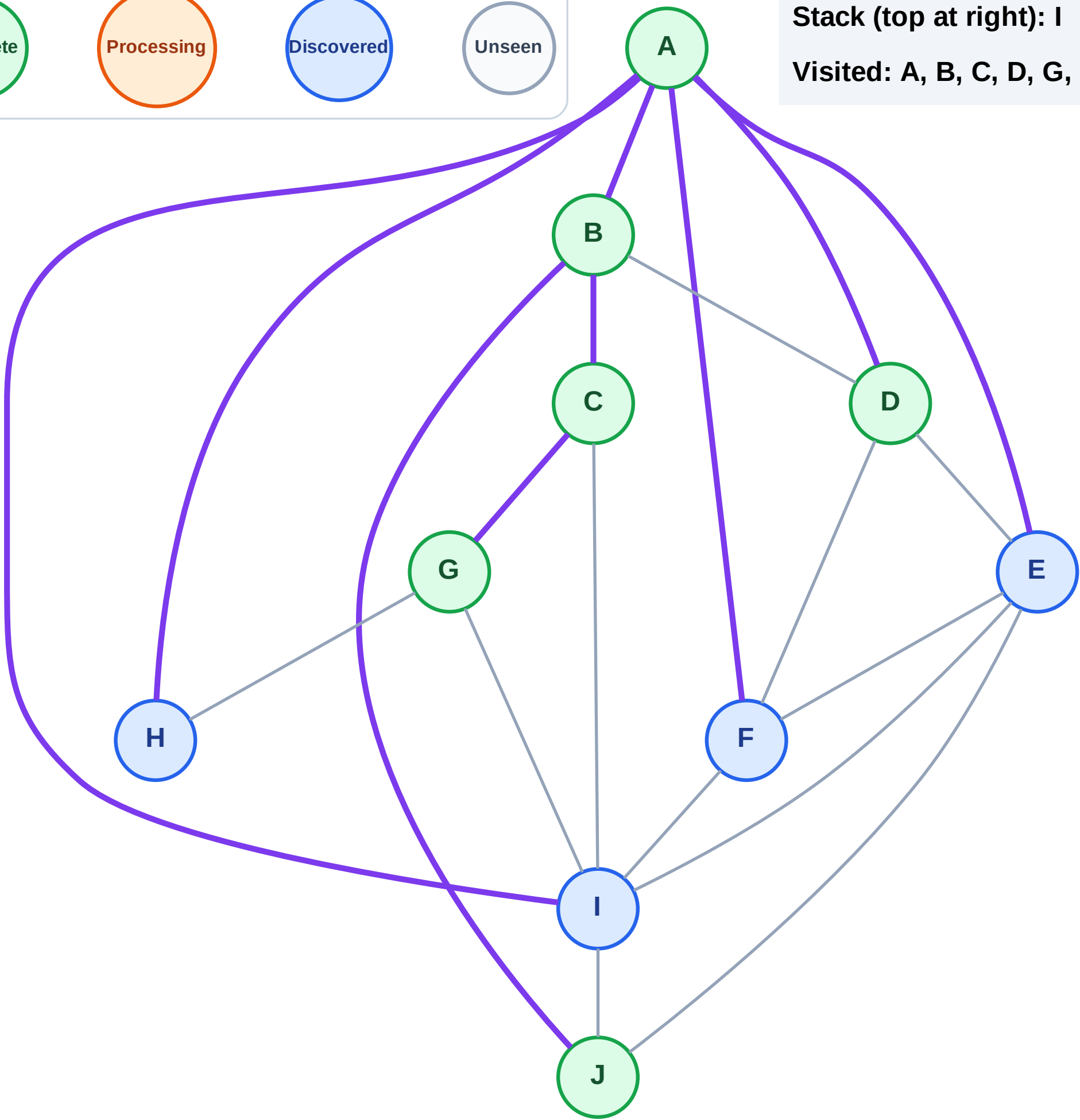
Step 13: No new neighbors from D

Legend



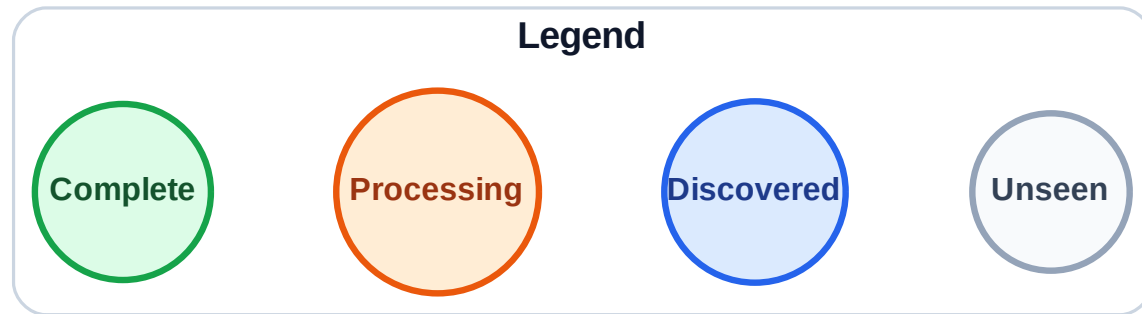
Stack (top at right): I | H | F | E

Visited: A, B, C, D, G, J



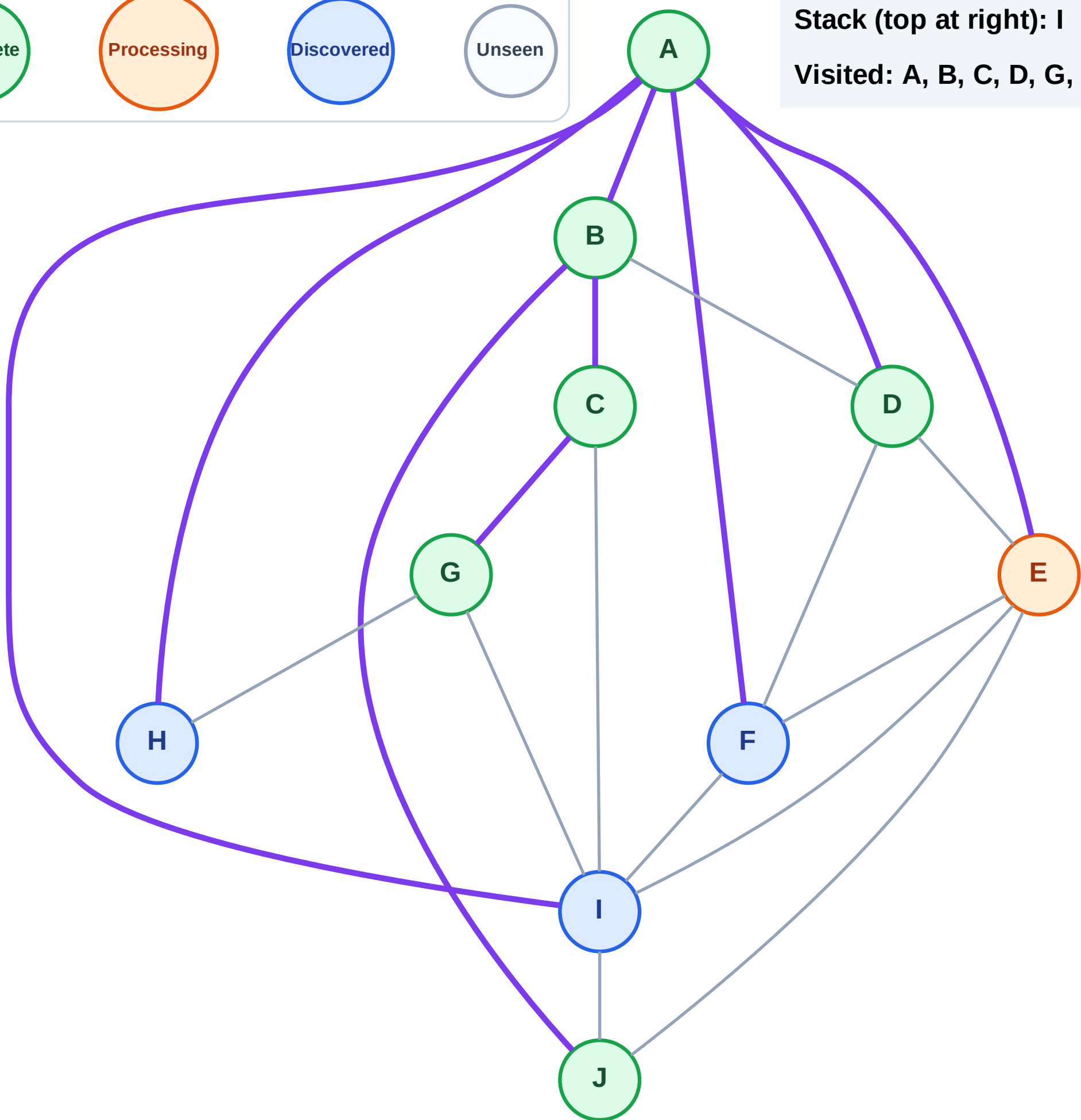
Step 14: Process node E

Step 14: Process node E



Stack (top at right): I | H | F

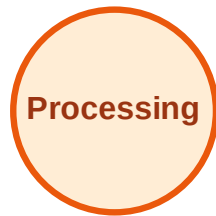
Visited: A, B, C, D, G, J



DFS Traversal

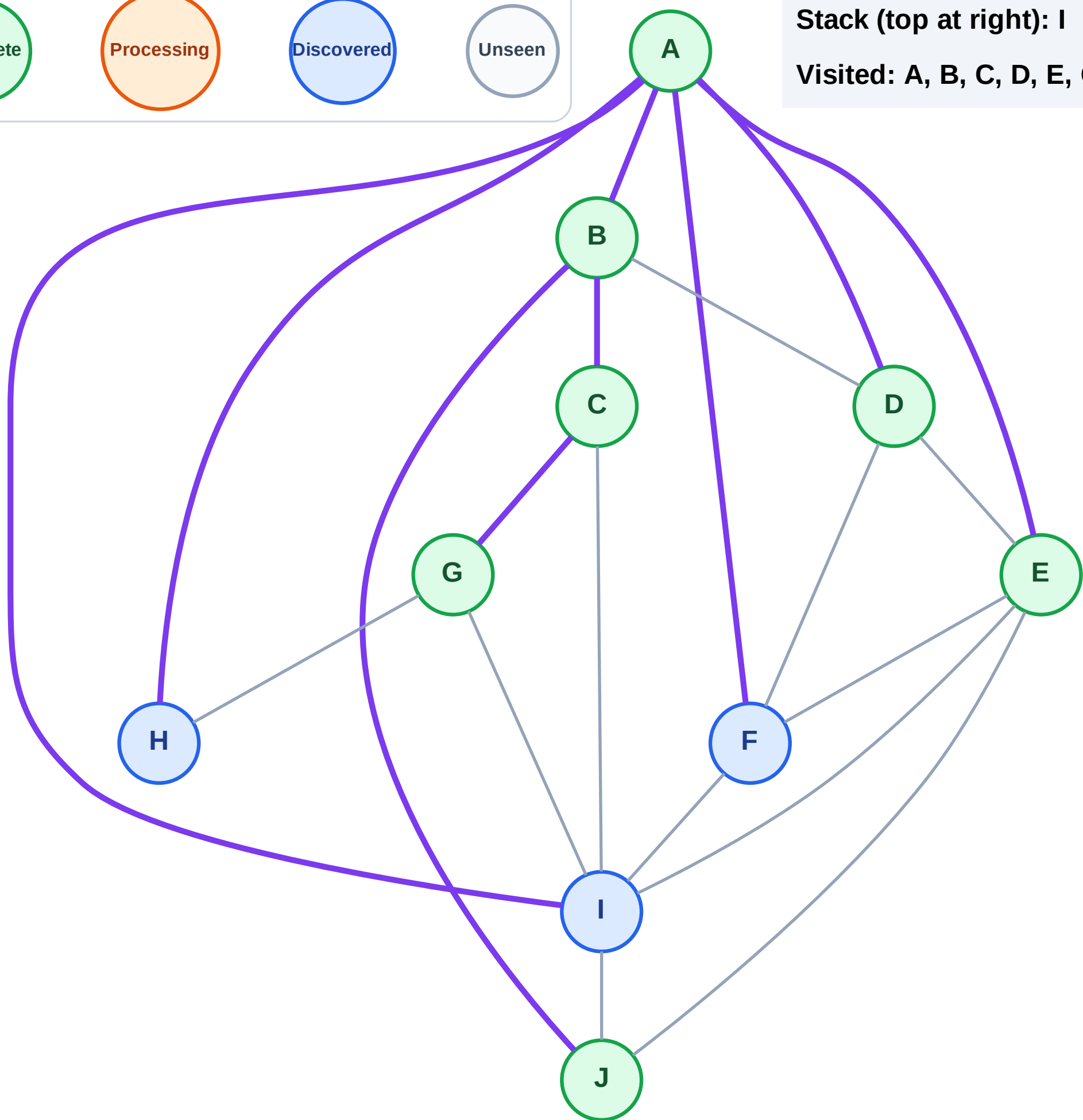
Step 15: No new neighbors from E

Legend



Stack (top at right): I | H | F

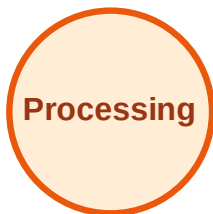
Visited: A, B, C, D, E, G, J



DFS Traversal

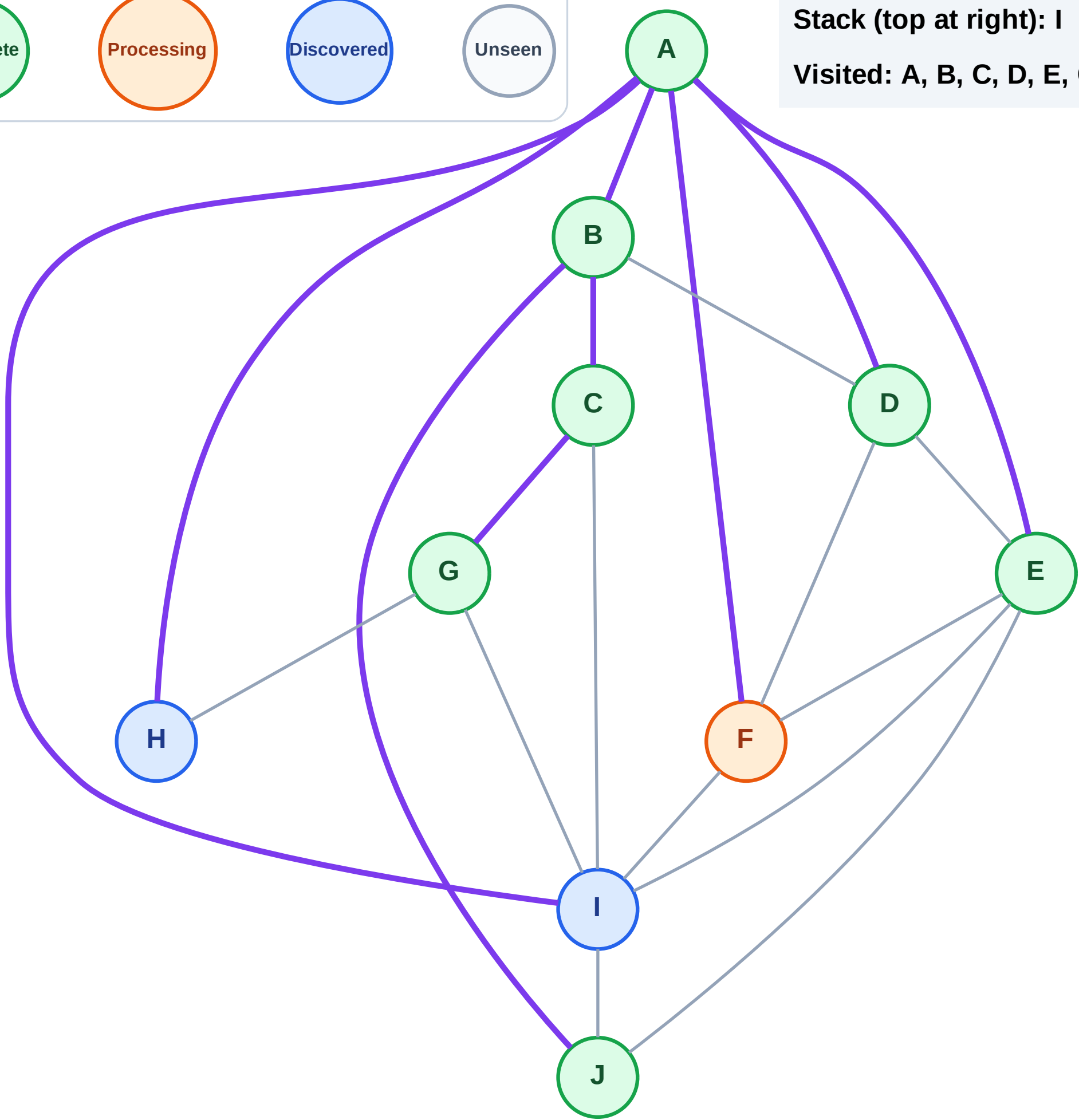
Step 16: Process node F

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, G, J

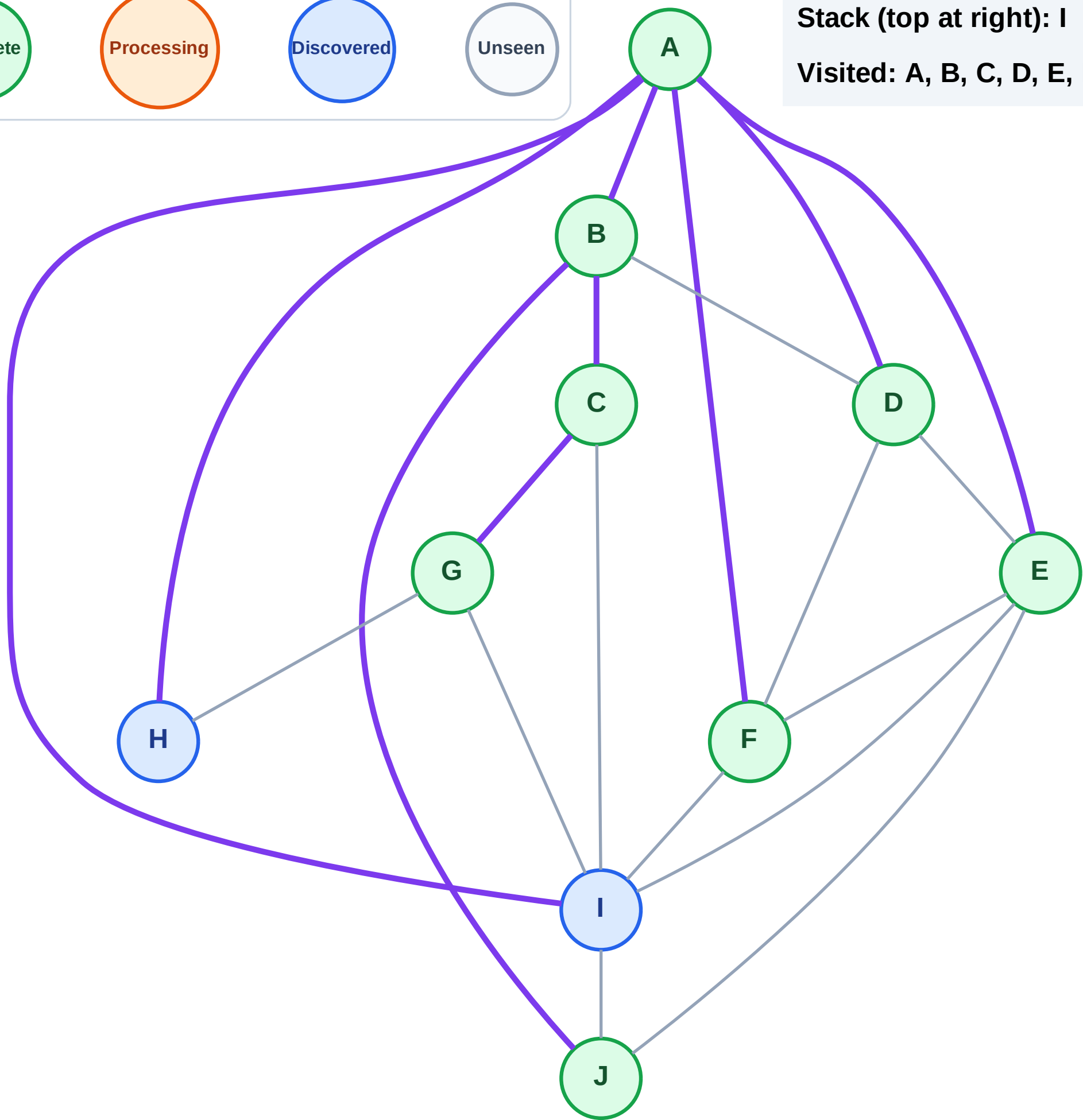


Step 17: No new neighbors from F

Legend



Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

Complete

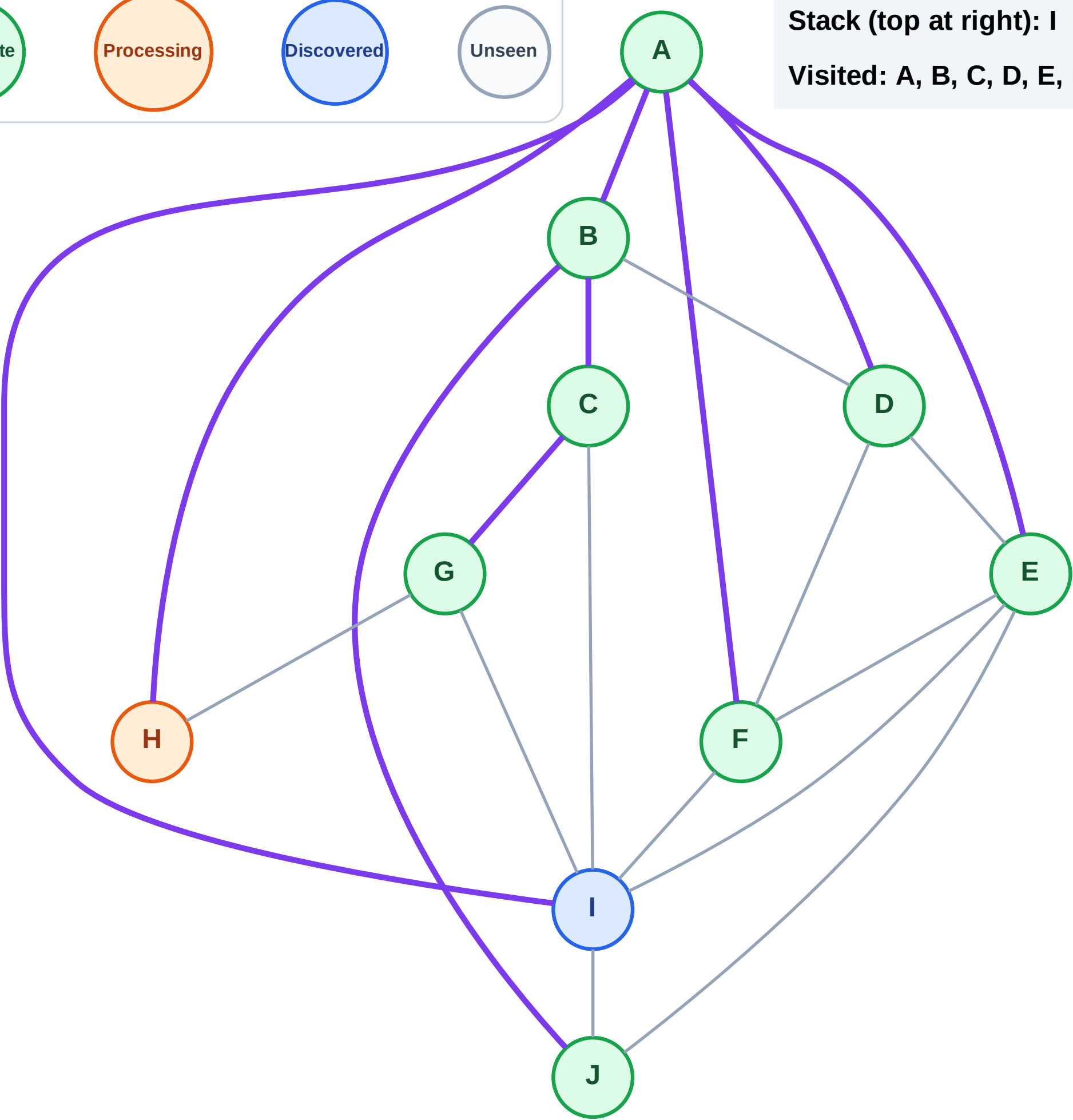
Processing

Discovered

Unseen

Stack (top at right): I

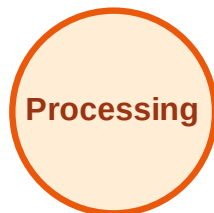
Visited: A, B, C, D, E, F, G, J



DFS Traversal

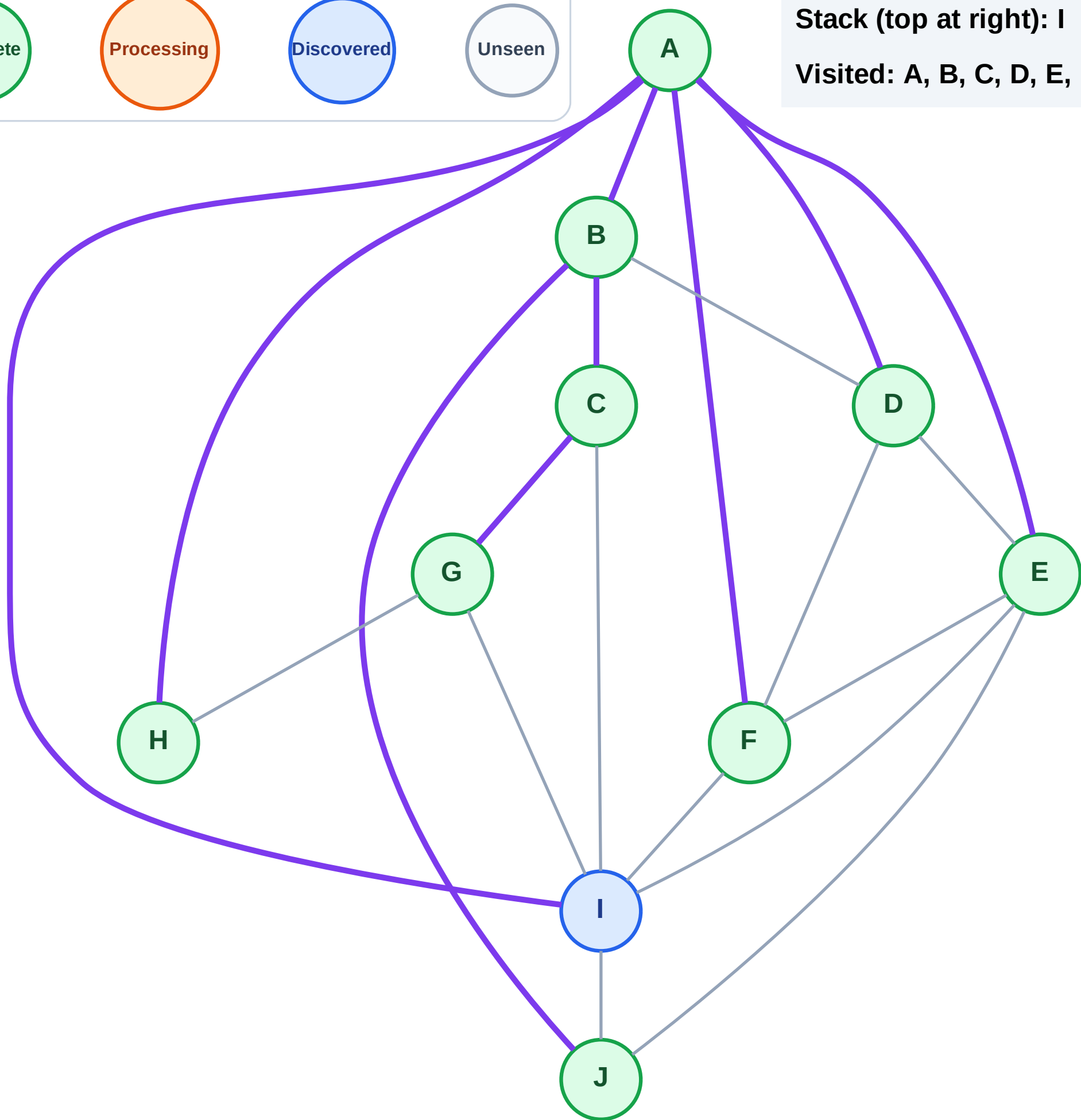
Step 19: No new neighbors from H

Legend



Stack (top at right): I

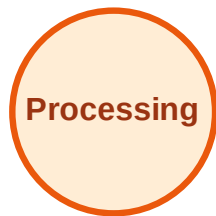
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

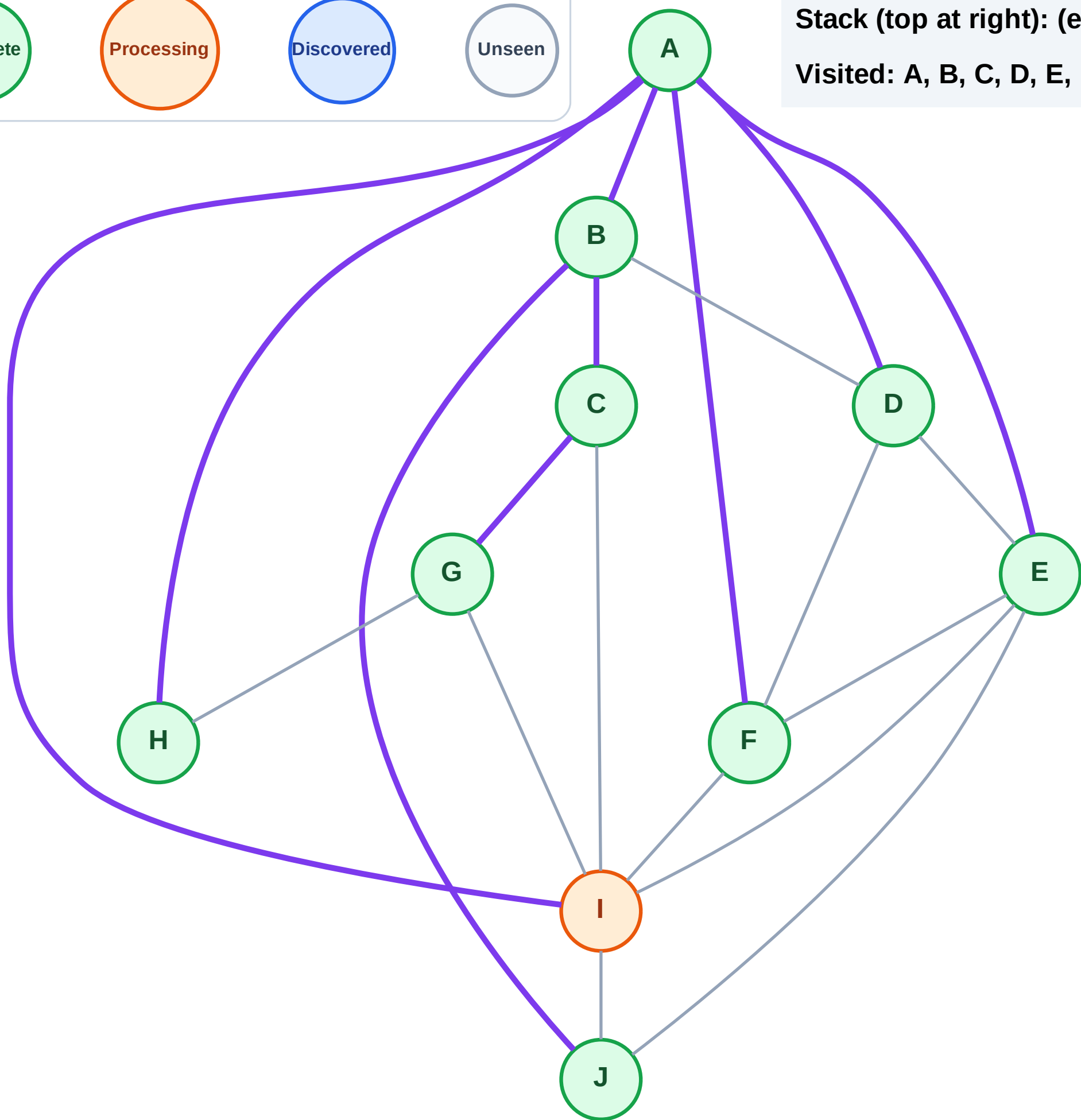
Step 20: Process node I

Legend



Stack (top at right): (empty)

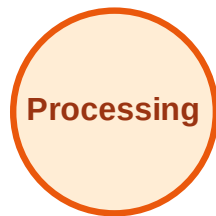
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

